

# BFS Traversal

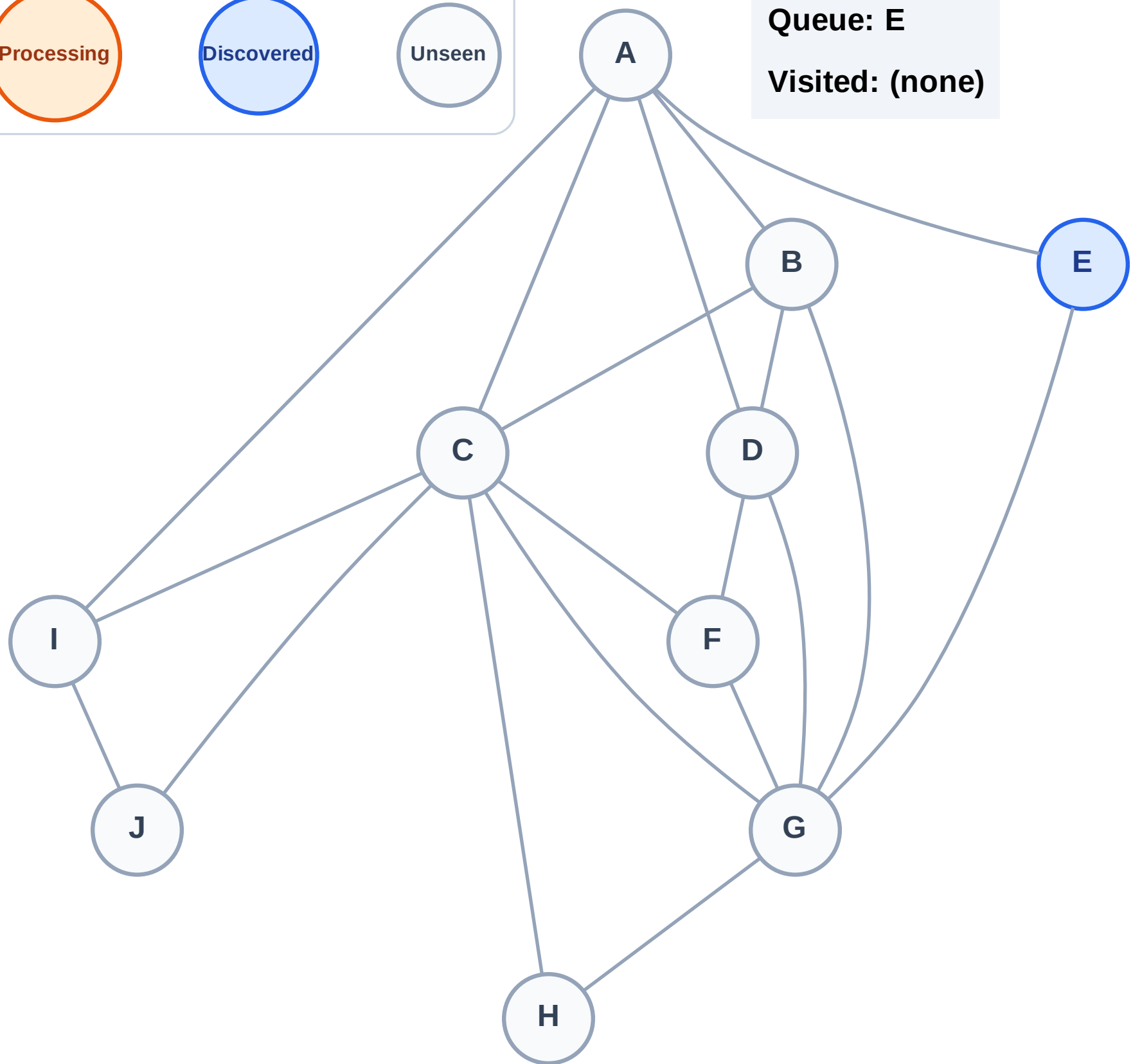
Step 1: Start at E

## Legend



Queue: E

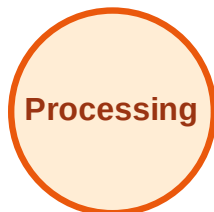
Visited: (none)



# BFS Traversal

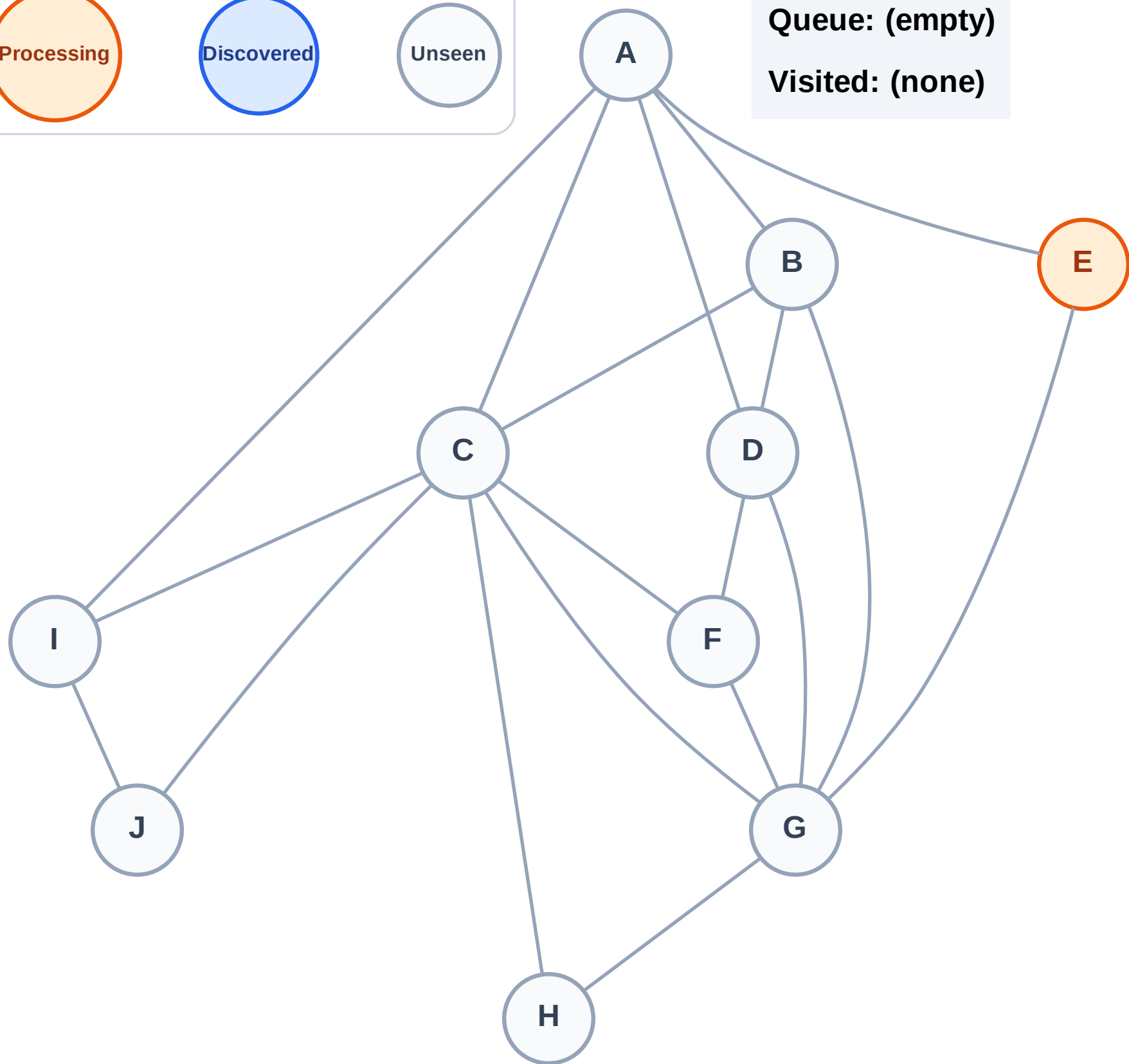
Step 2: Process node E

## Legend



Queue: (empty)

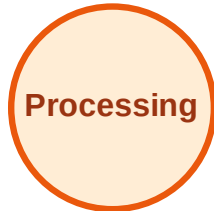
Visited: (none)



# BFS Traversal

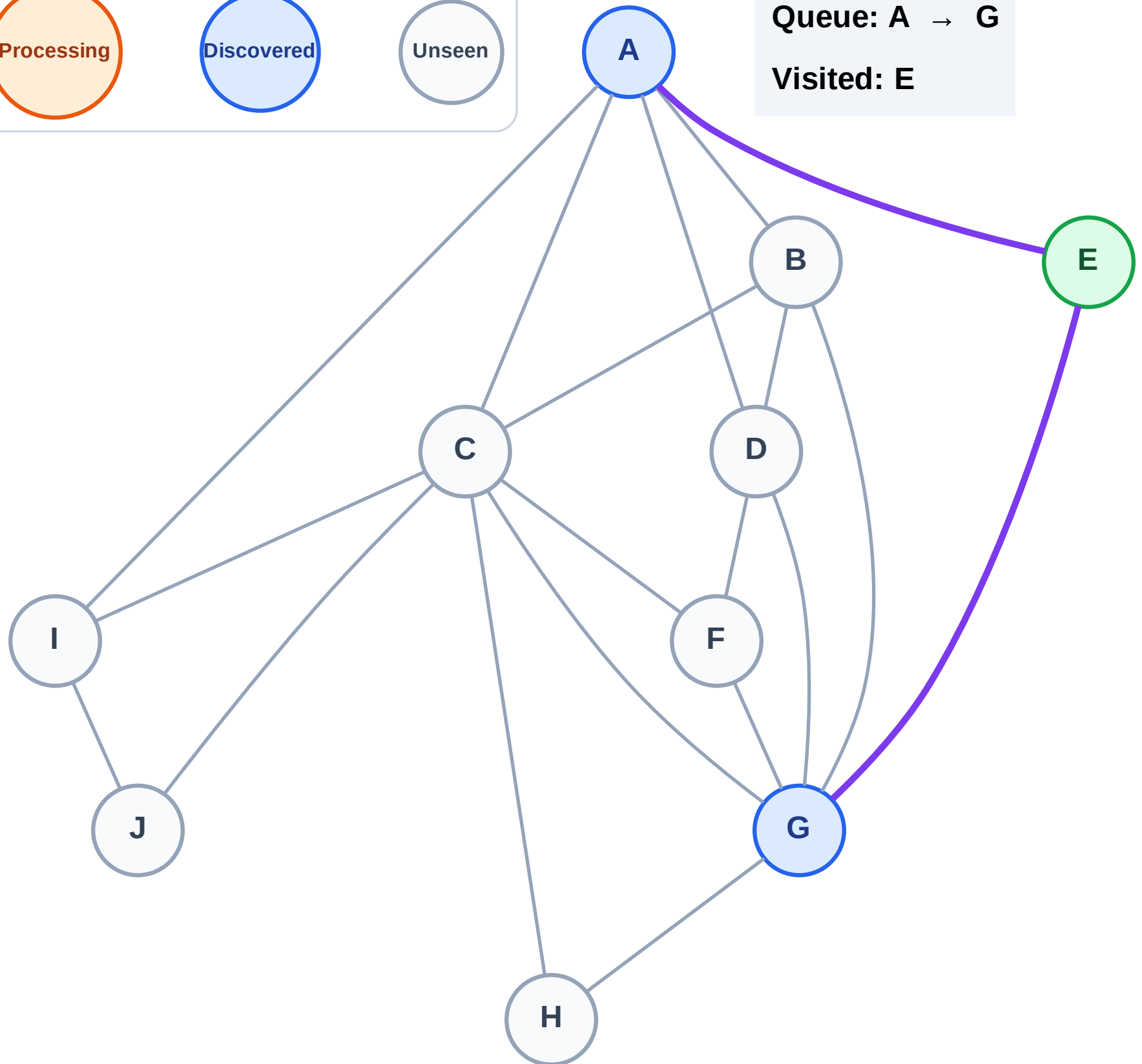
Step 3: Discover A, G from E

## Legend



Queue: A → G

Visited: E



# BFS Traversal

Step 4: Process node A

Legend

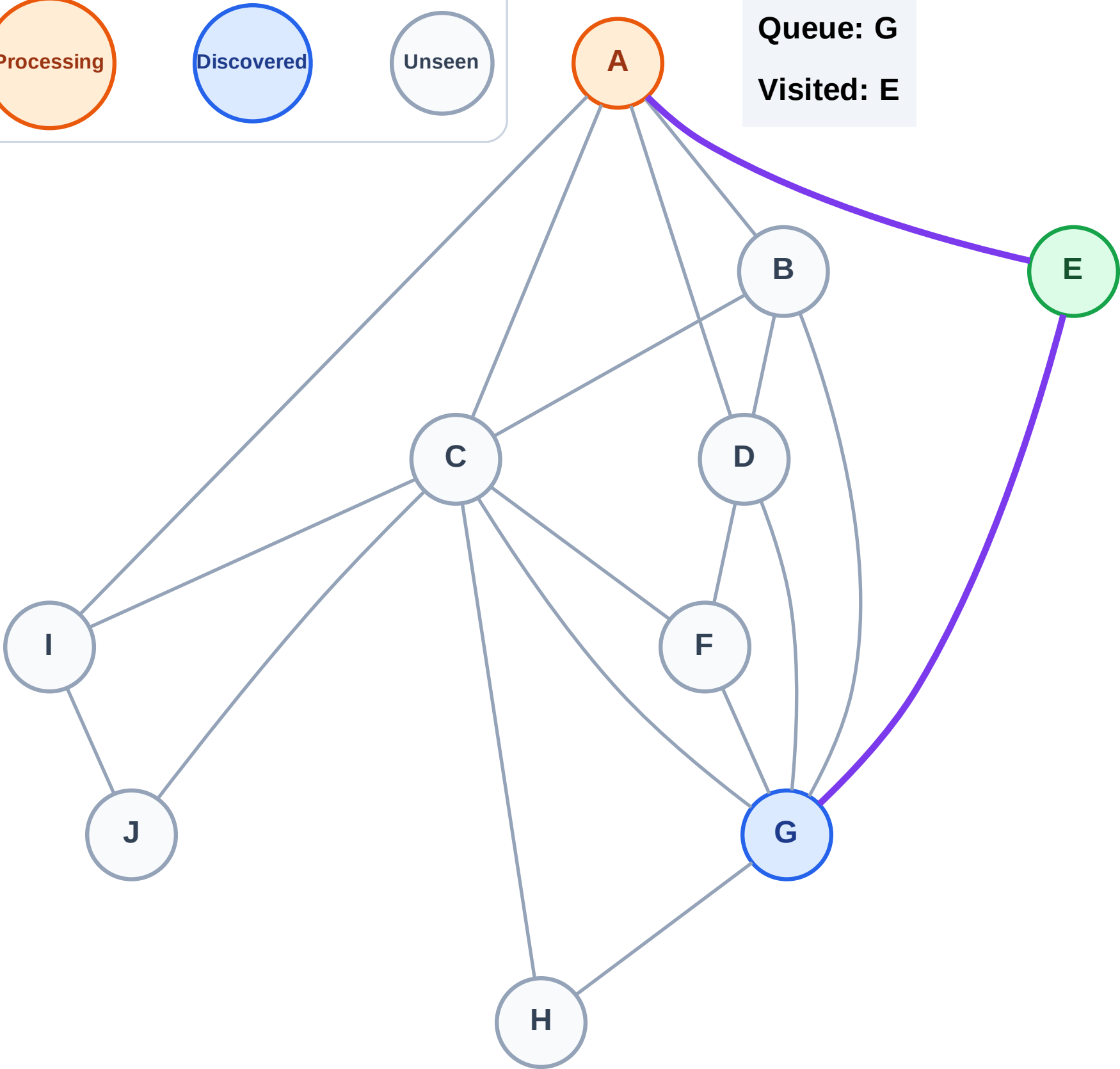
Complete

Processing

Discovered

Unseen

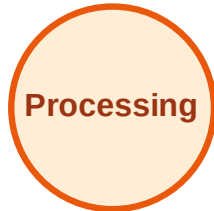
Queue: G  
Visited: E



# BFS Traversal

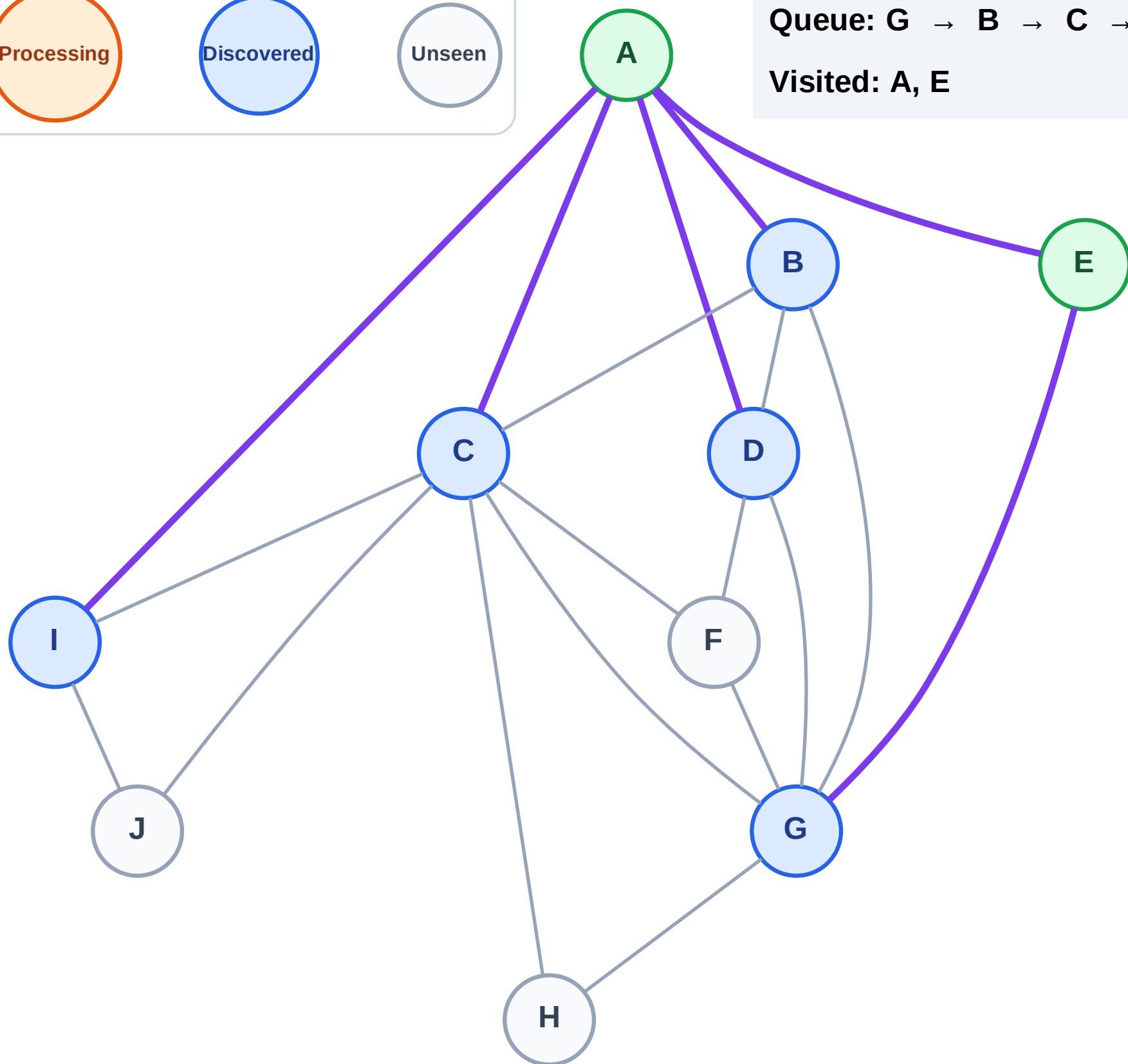
Step 5: Discover B, C, D, I from A

## Legend



Queue: G → B → C → D → I

Visited: A, E



# BFS Traversal

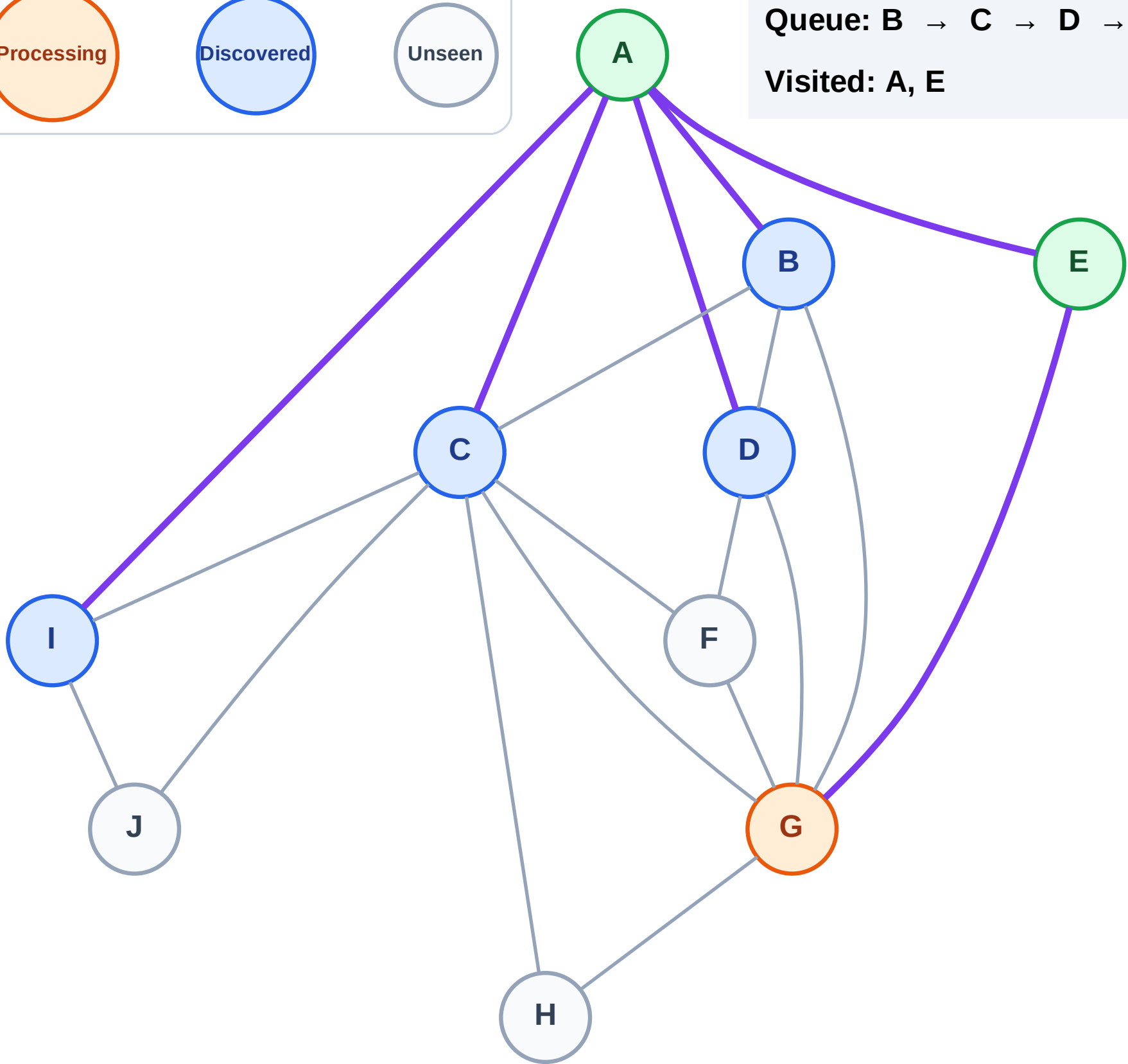
Step 6: Process node G

## Legend



Queue: B → C → D → I

Visited: A, E



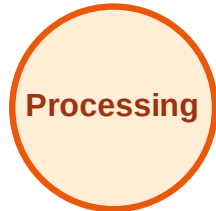
# BFS Traversal

Step 7: Discover F, H from G

## Legend



Complete



Processing

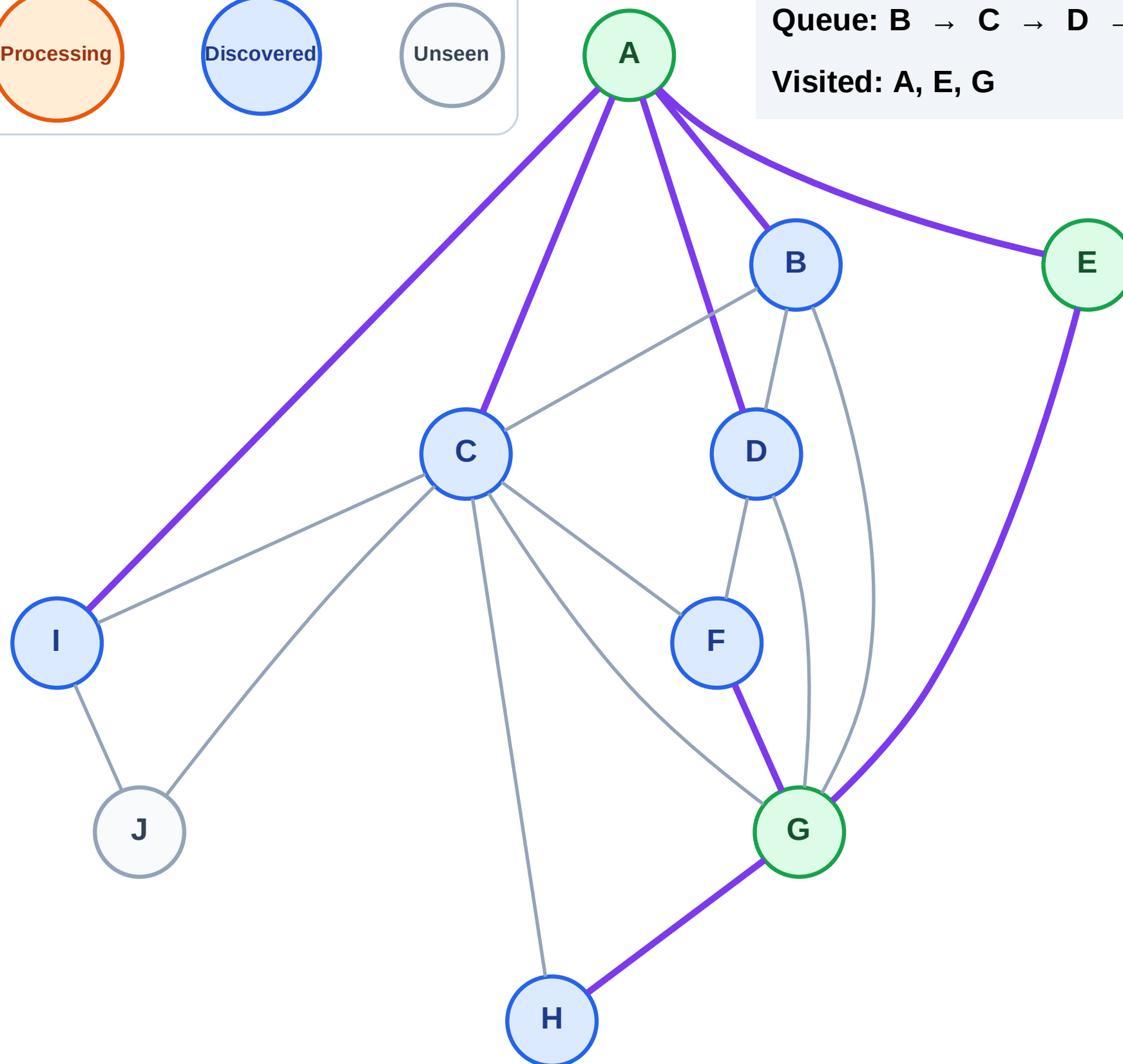


Discovered



Unseen

Queue: B → C → D → I → F → H  
Visited: A, E, G





# BFS Traversal

Step 8: Process node B

Legend

Complete

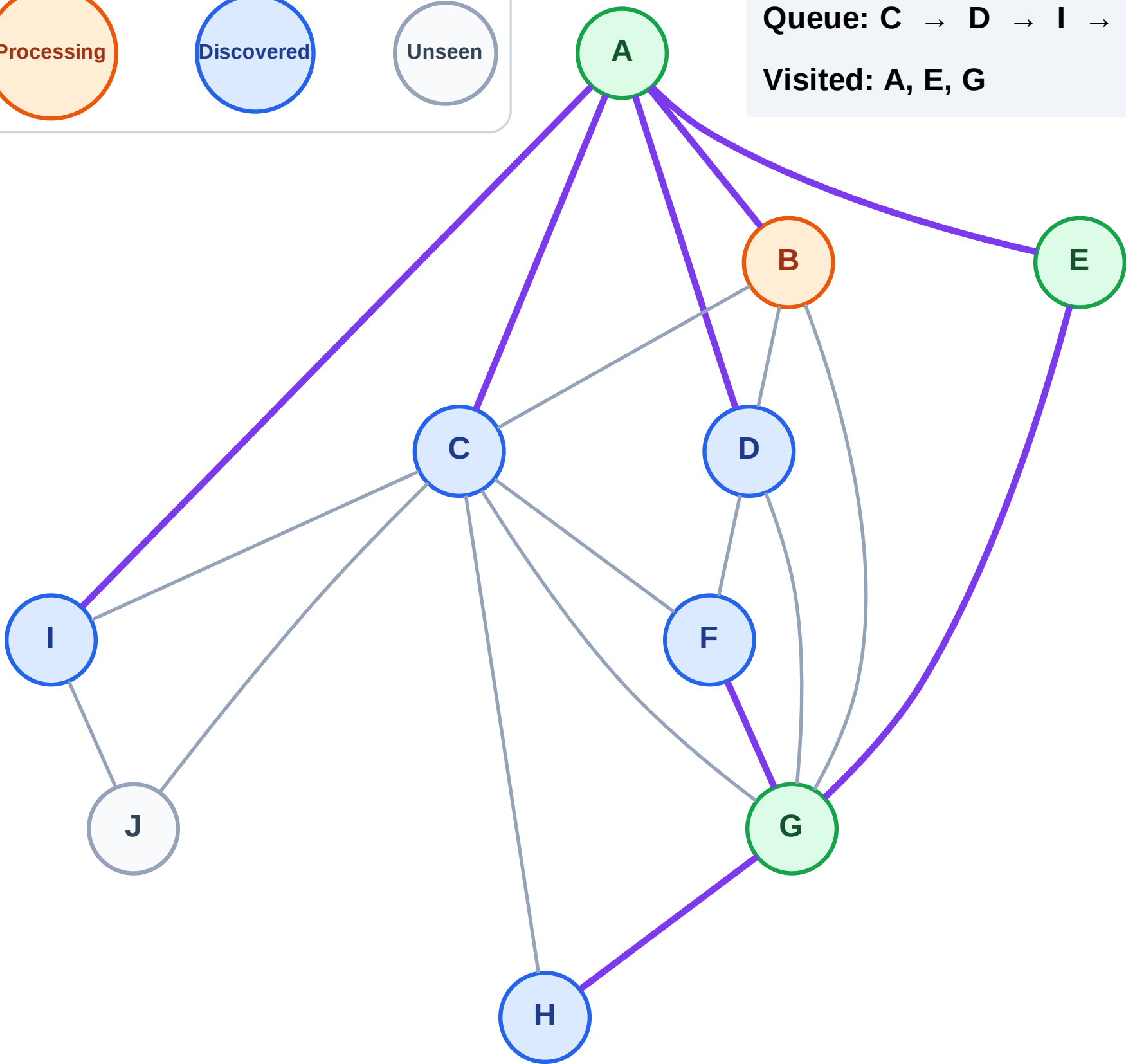
Processing

Discovered

Unseen

Queue: C → D → I → F → H

Visited: A, E, G

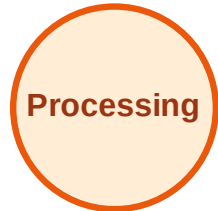




# BFS Traversal

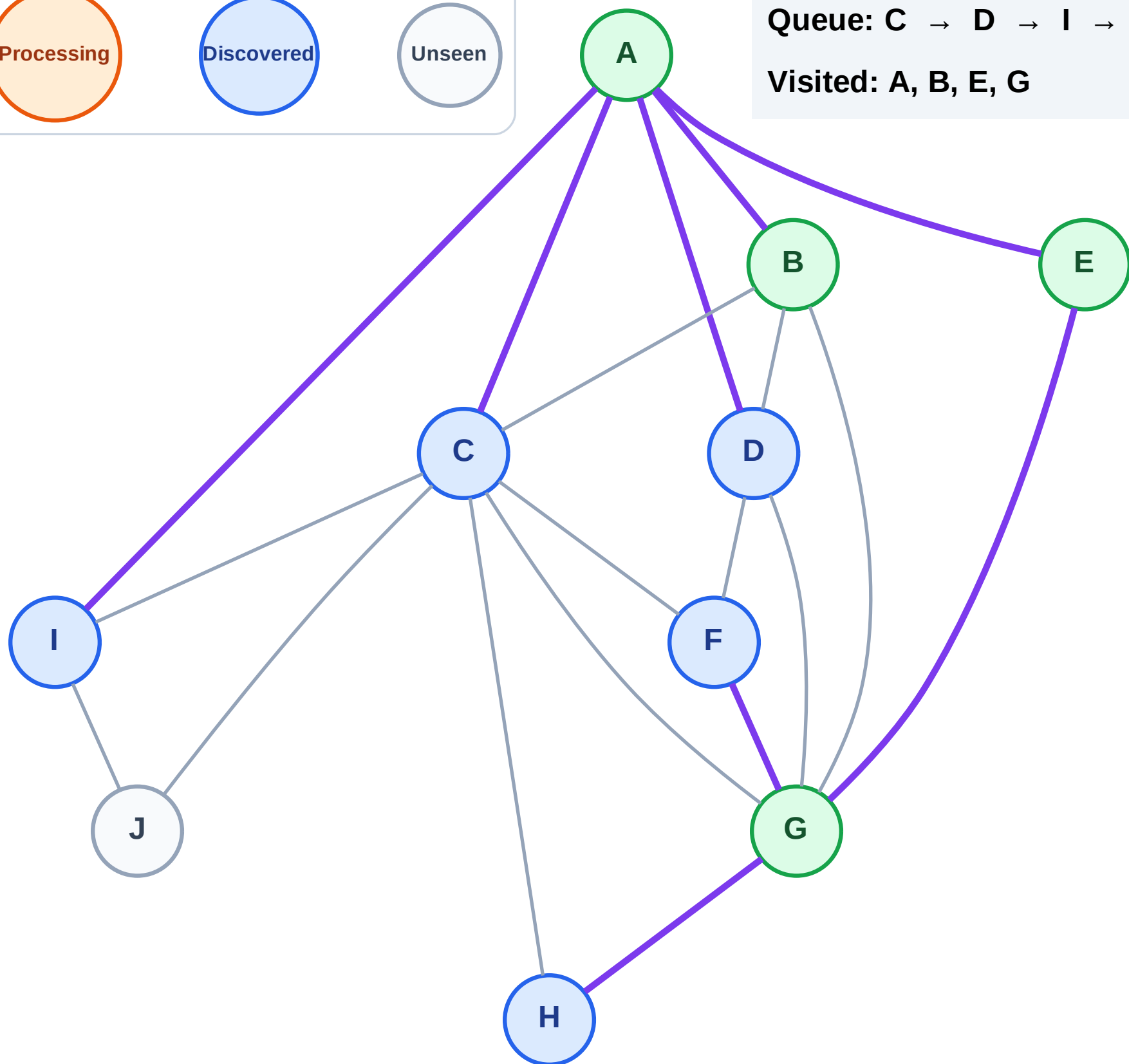
Step 9: No new neighbors from B

## Legend



Queue: C → D → I → F → H

Visited: A, B, E, G



# BFS Traversal

Step 10: Process node C

Legend

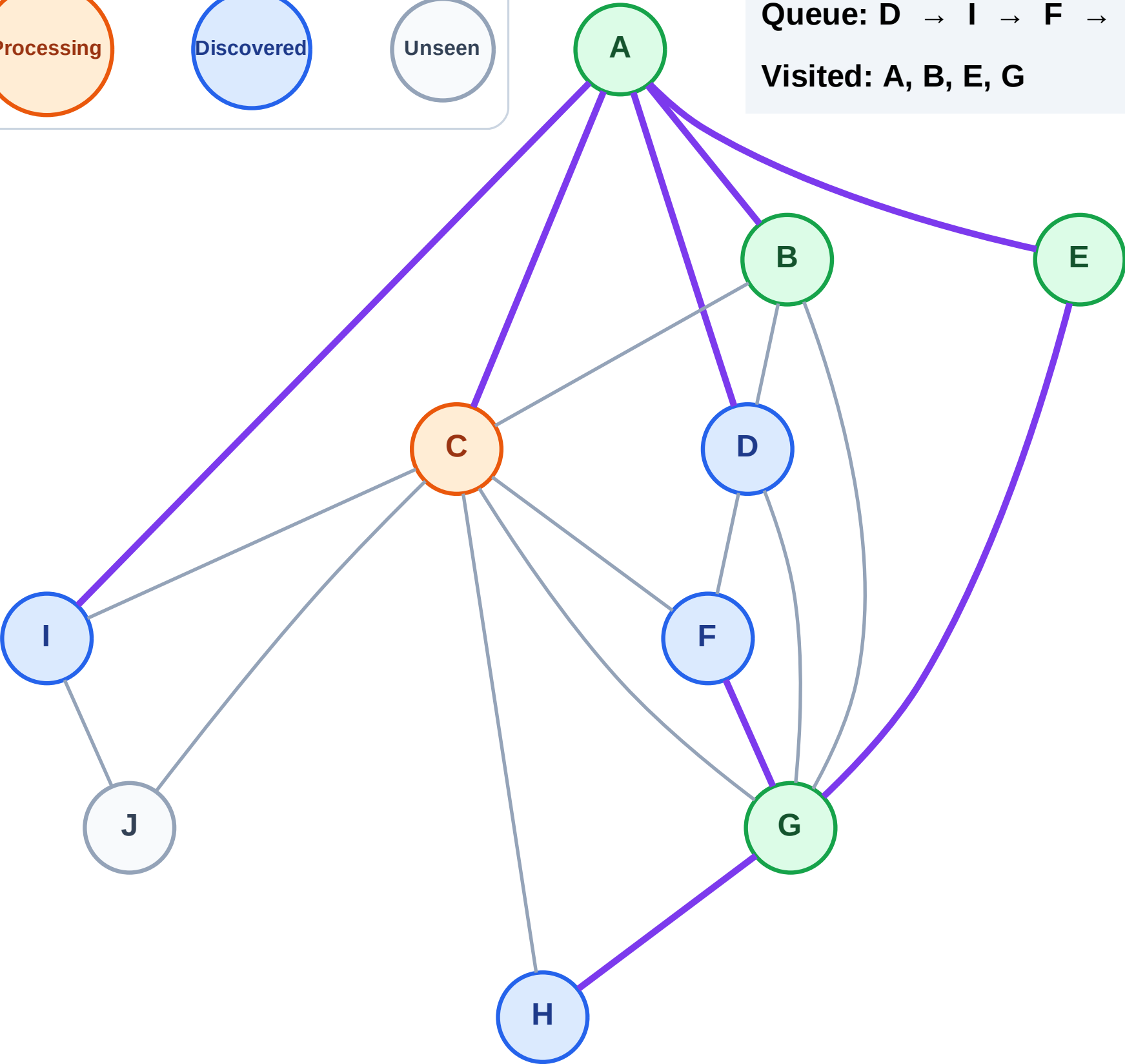
Complete

Processing

Discovered

Unseen

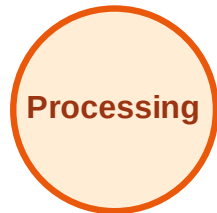
Queue: D → I → F → H  
Visited: A, B, E, G



# BFS Traversal

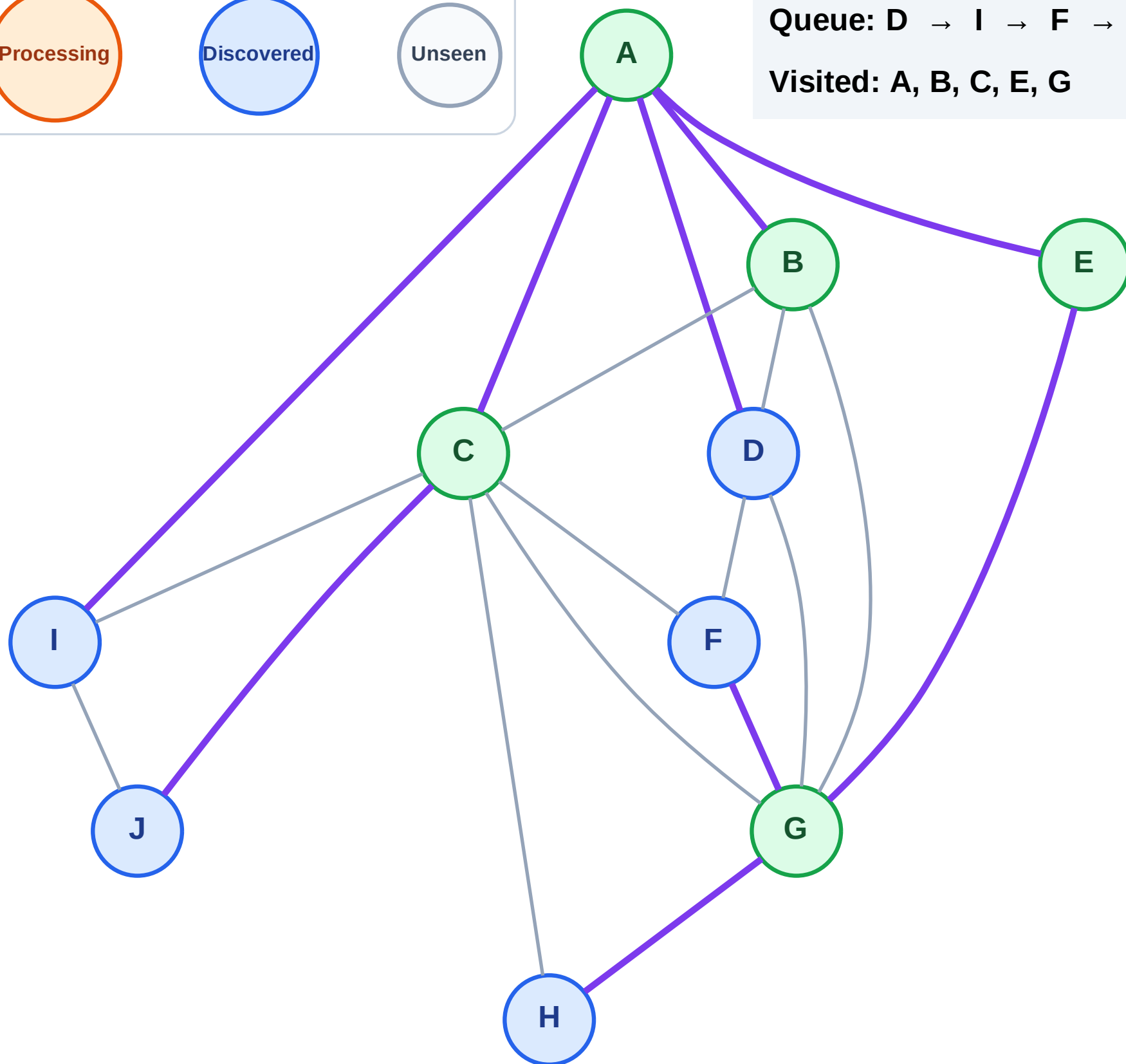
Step 11: Discover J from C

## Legend



Queue: D → I → F → H → J

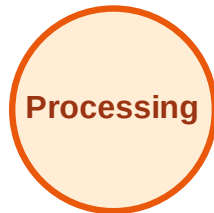
Visited: A, B, C, E, G



# BFS Traversal

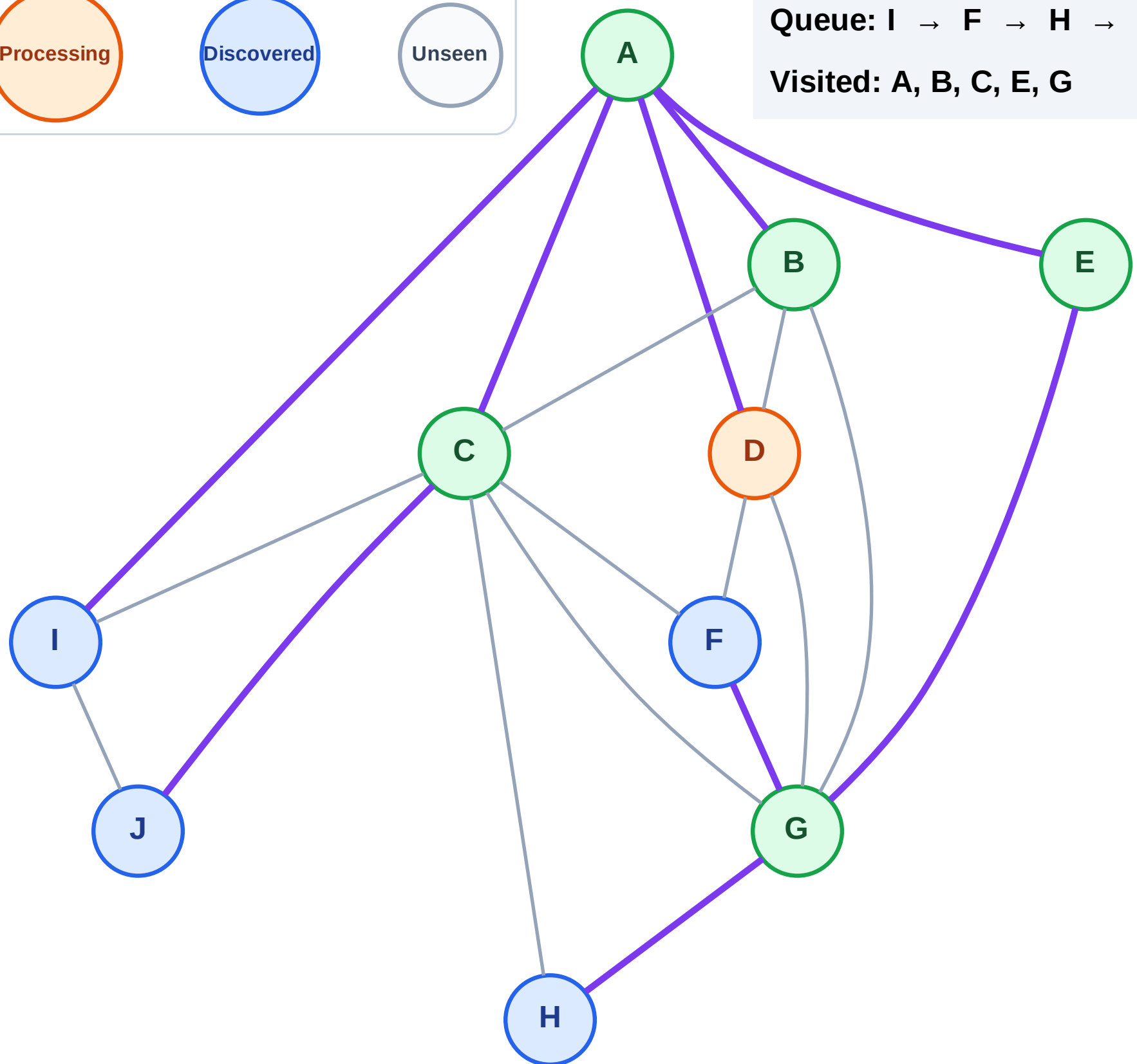
Step 12: Process node D

## Legend



Queue: I → F → H → J

Visited: A, B, C, E, G



# BFS Traversal

Step 13: No new neighbors from D

Legend

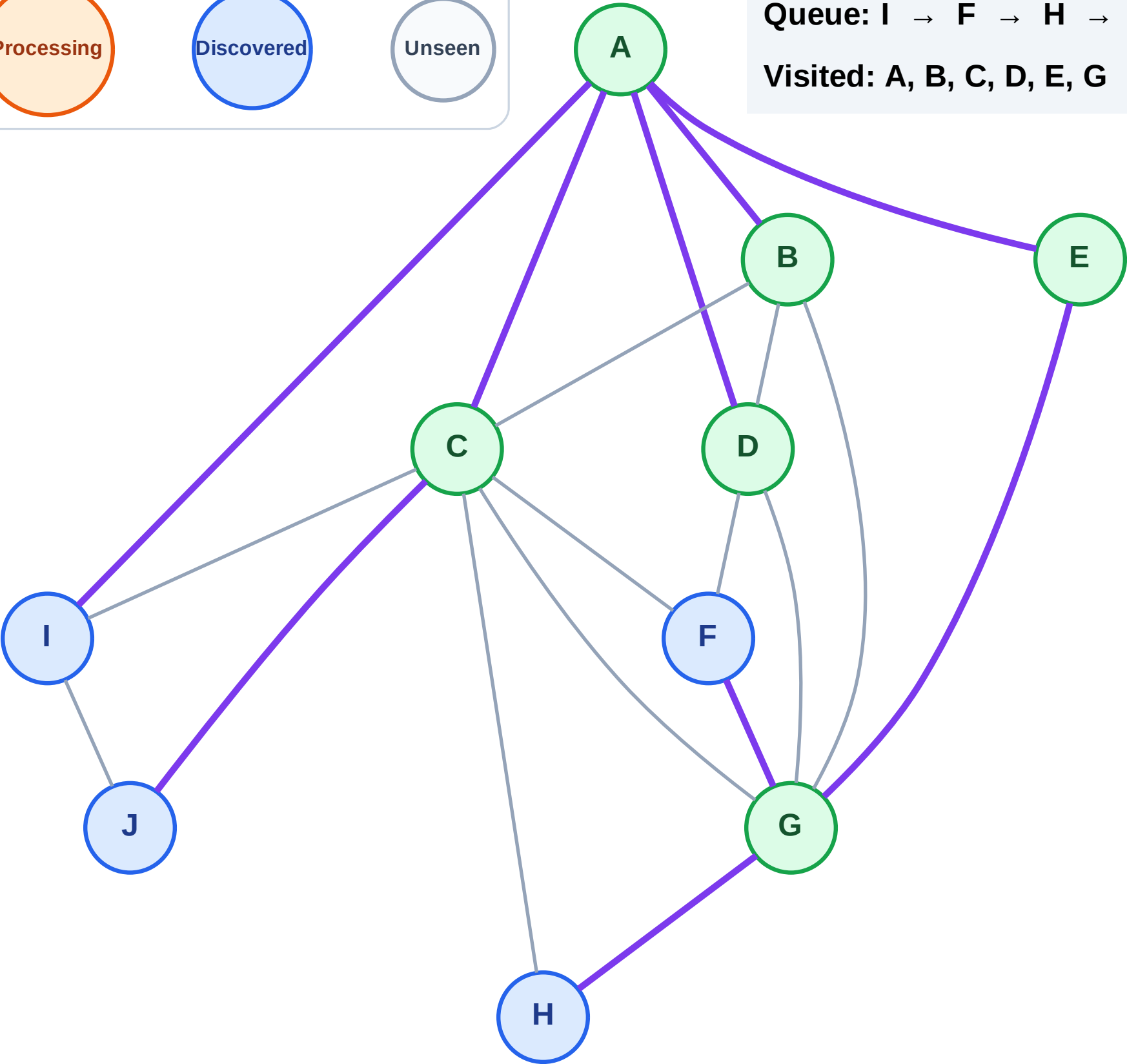
Complete

Processing

Discovered

Unseen

Queue: I → F → H → J  
Visited: A, B, C, D, E, G



# BFS Traversal

Step 14: Process node I

Legend

Complete

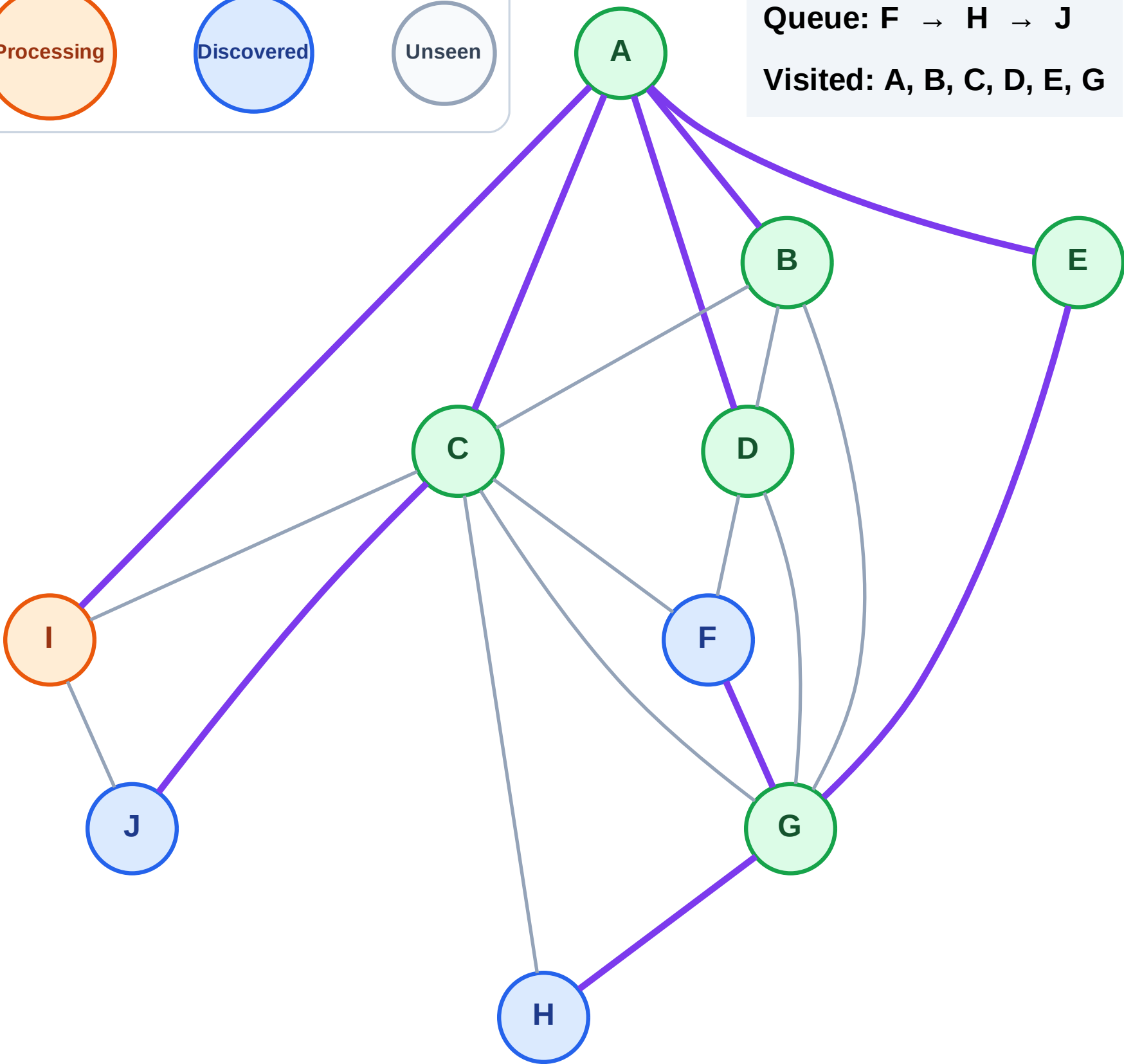
Processing

Discovered

Unseen

Queue: F → H → J

Visited: A, B, C, D, E, G



# BFS Traversal

Step 15: No new neighbors from I

Legend

Complete

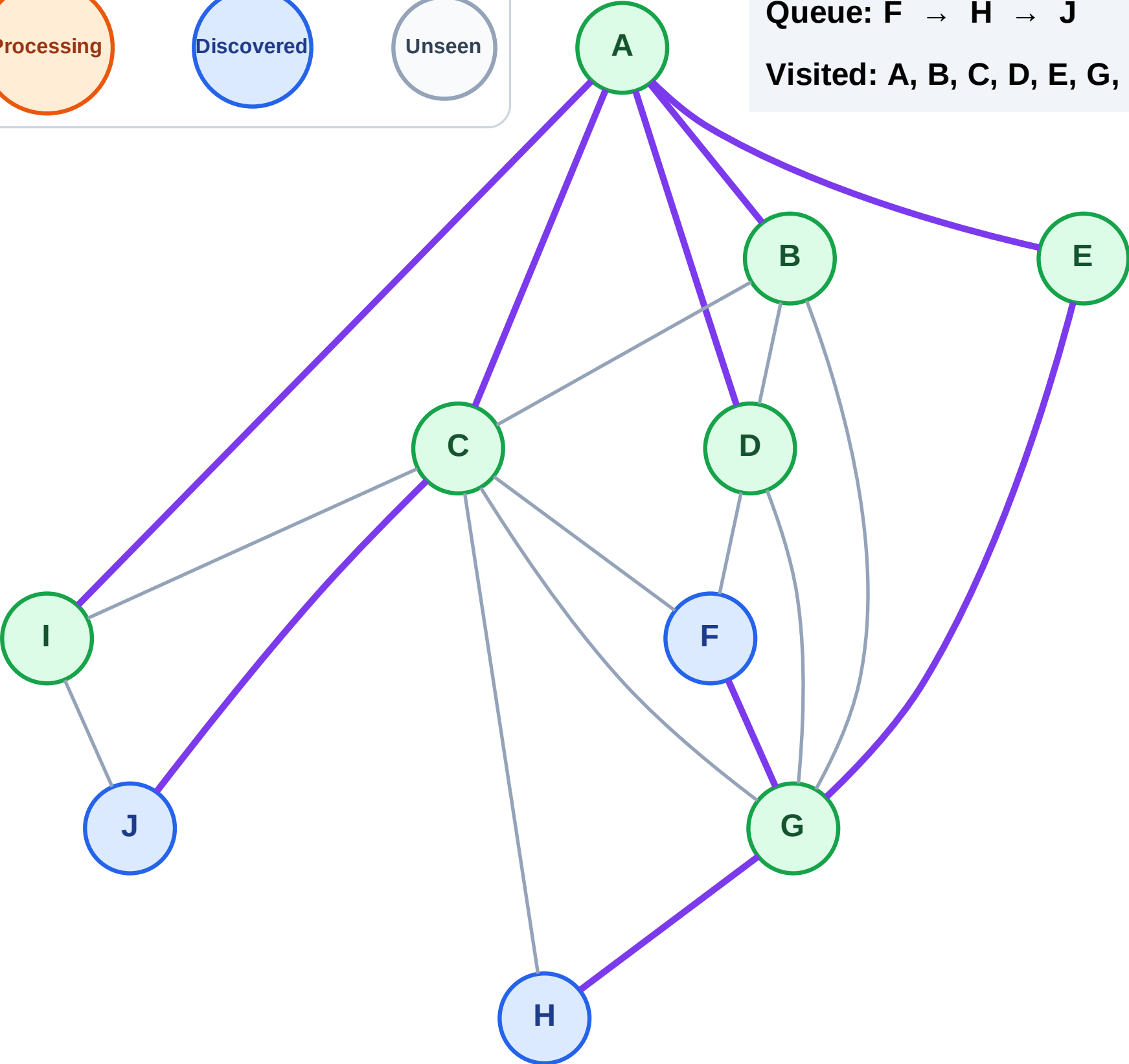
Processing

Discovered

Unseen

Queue: F → H → J

Visited: A, B, C, D, E, G, I





# BFS Traversal

Step 16: Process node F

Legend

Complete

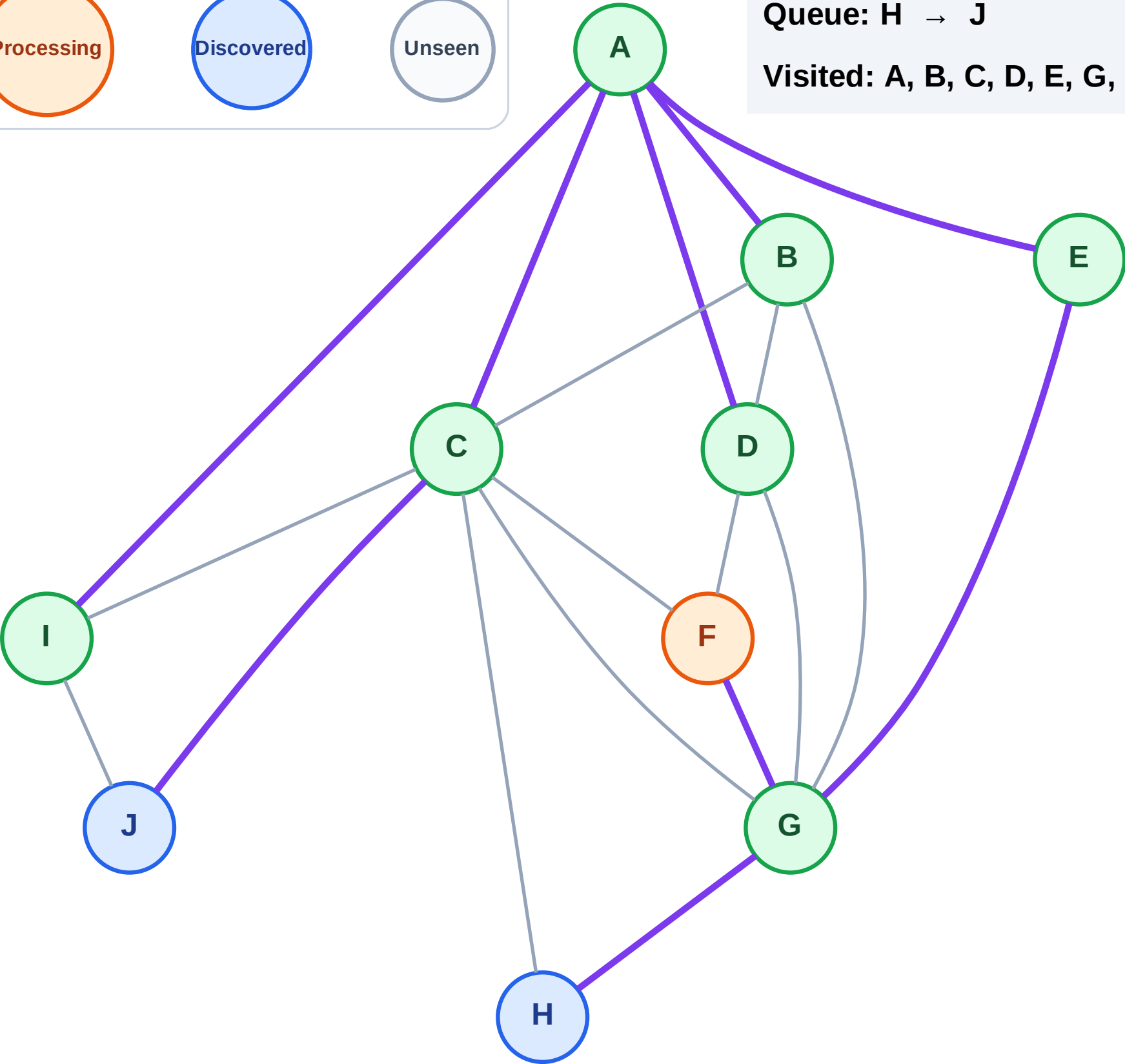
Processing

Discovered

Unseen

Queue: H → J

Visited: A, B, C, D, E, G, I

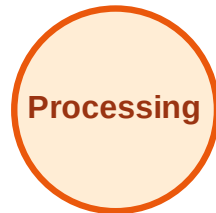




# BFS Traversal

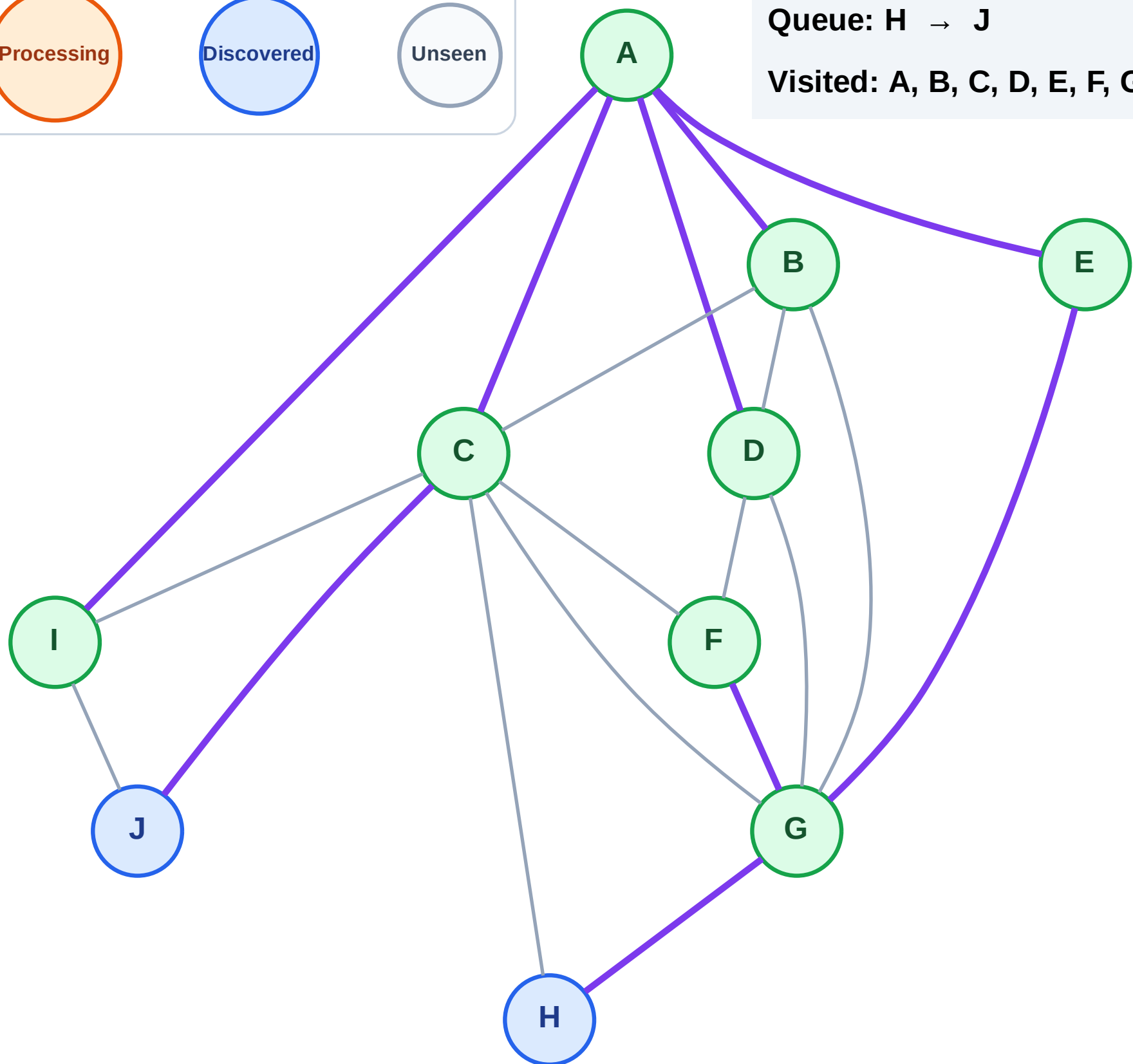
Step 17: No new neighbors from F

## Legend



Queue: H → J

Visited: A, B, C, D, E, F, G, I



# BFS Traversal

Step 18: Process node H

Legend

Complete

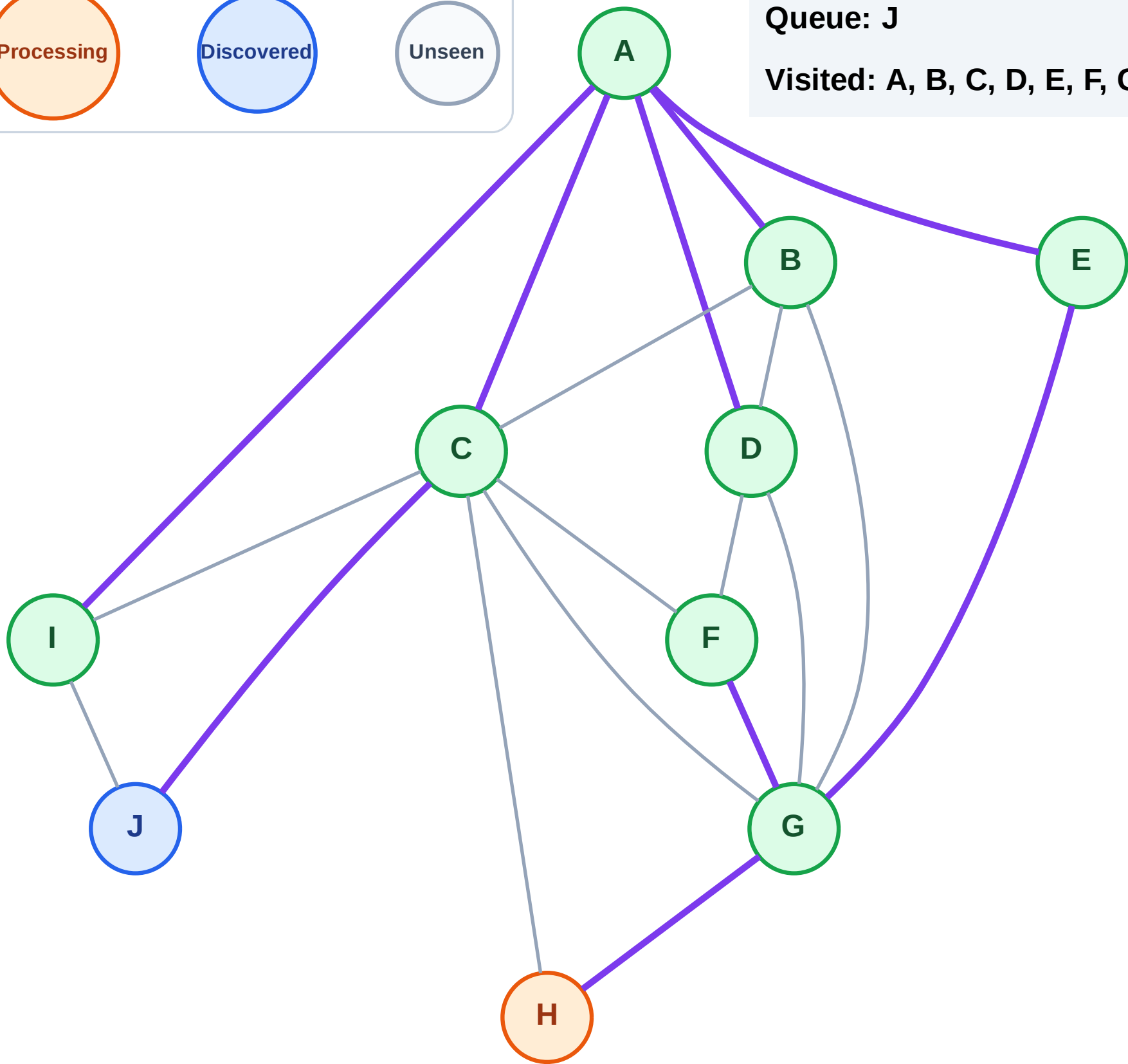
Processing

Discovered

Unseen

Queue: J

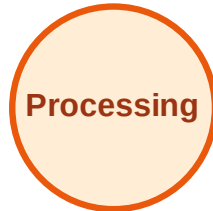
Visited: A, B, C, D, E, F, G, I



# BFS Traversal

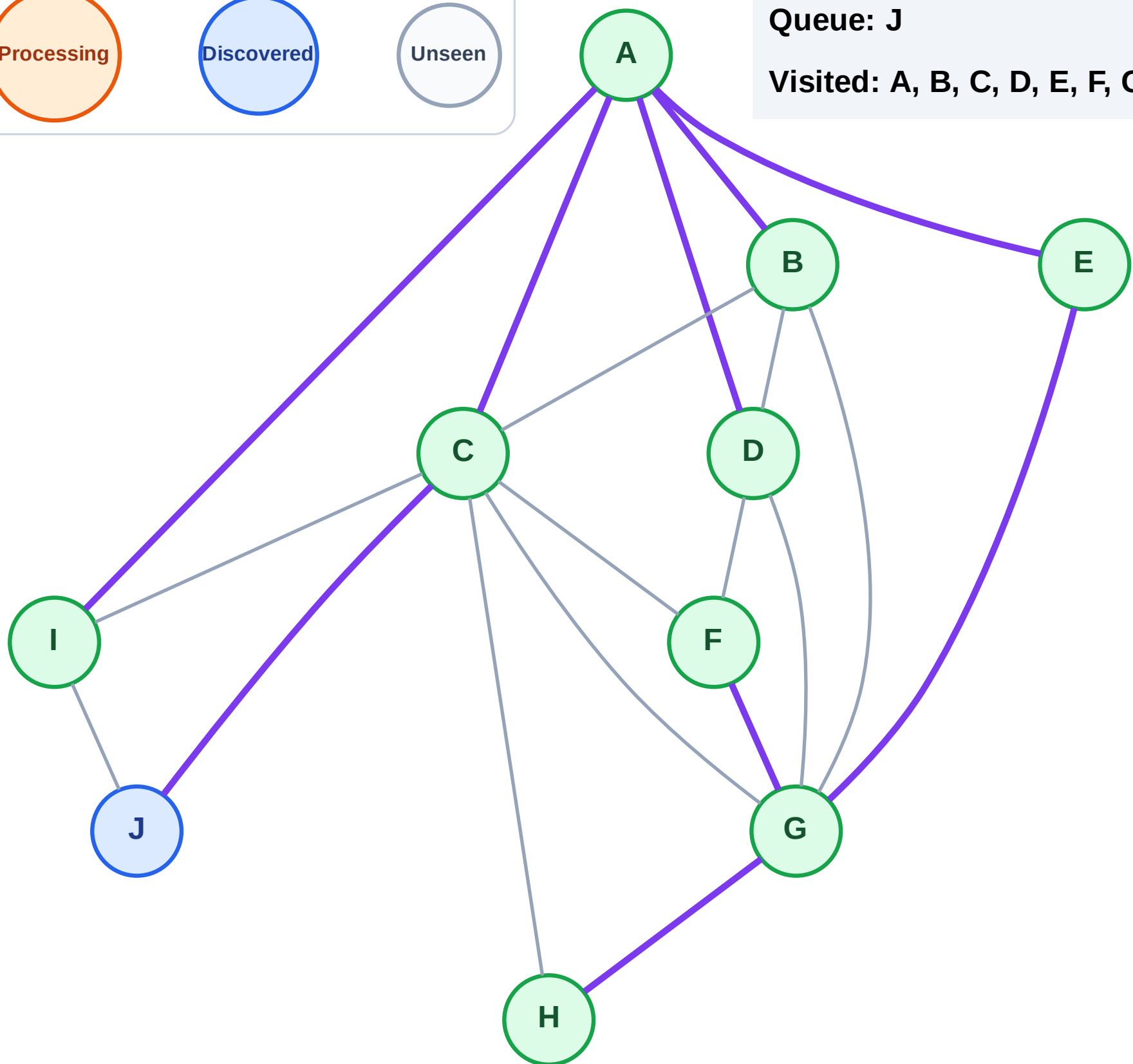
Step 19: No new neighbors from H

## Legend



Queue: J

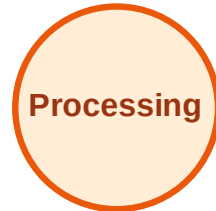
Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

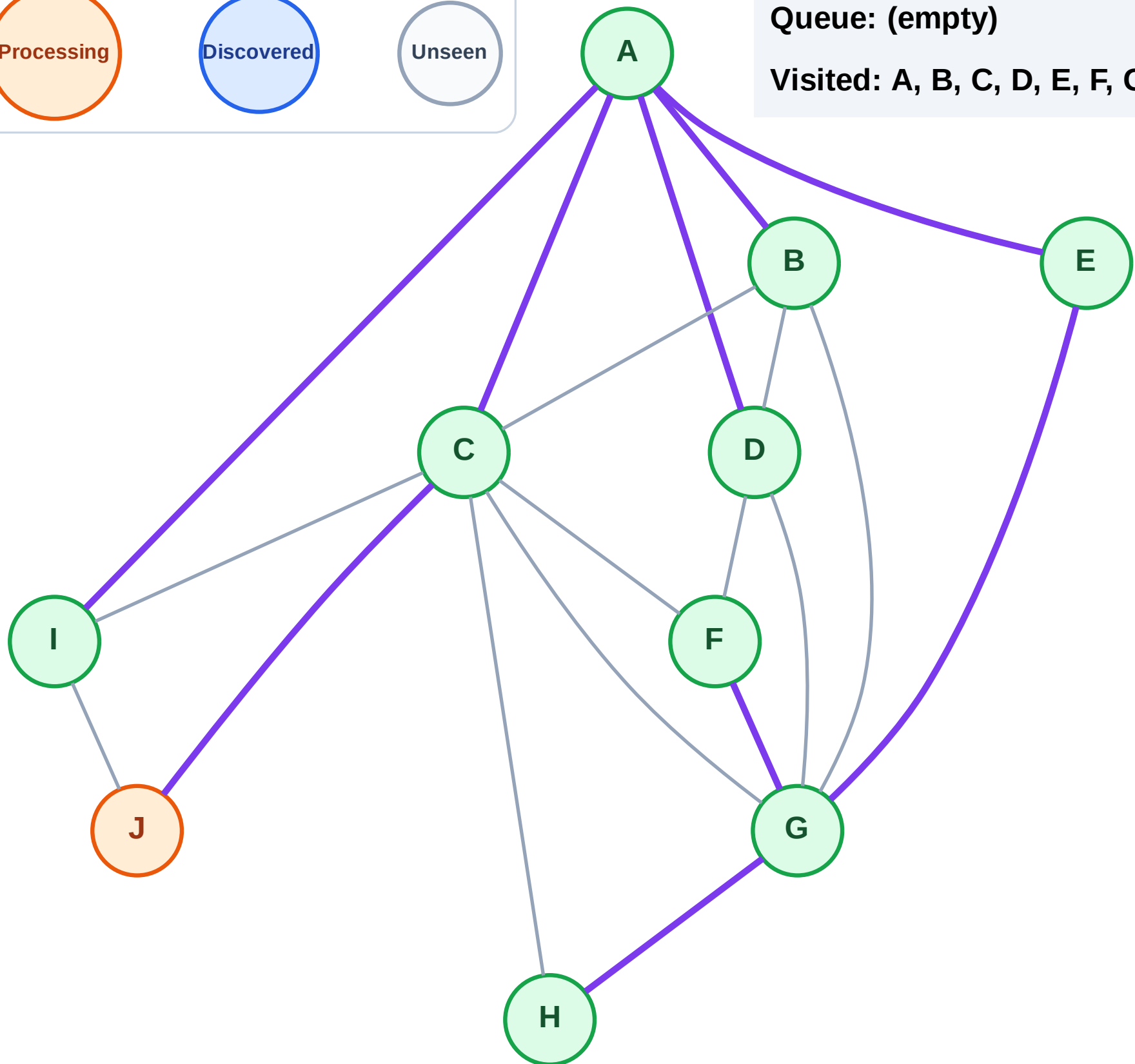
Step 20: Process node J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I



# BFS Traversal

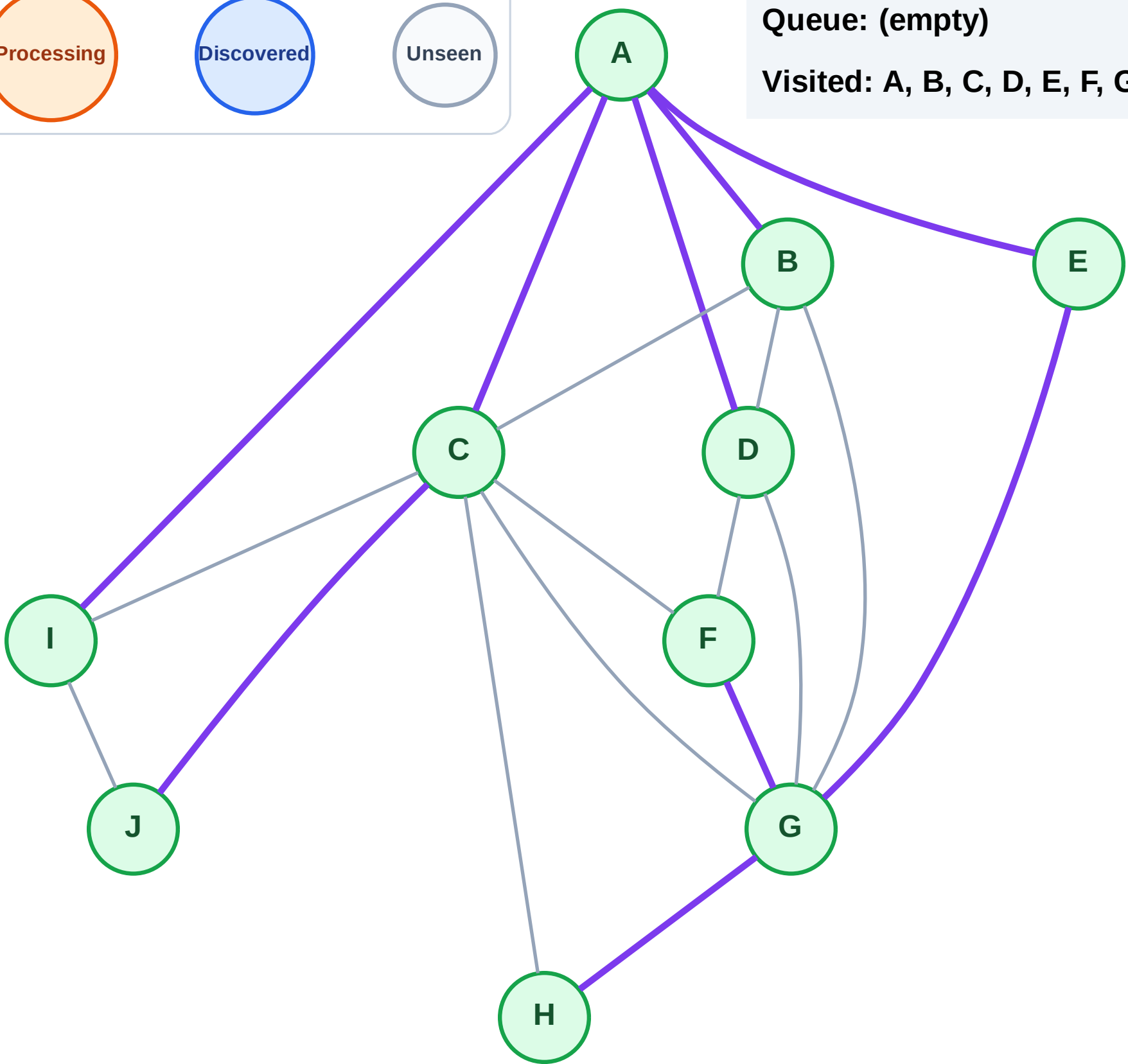
Step 21: No new neighbors from J

## Legend



Queue: (empty)

Visited: A, B, C, D, E, F, G, H, I, J



# DFS Traversal

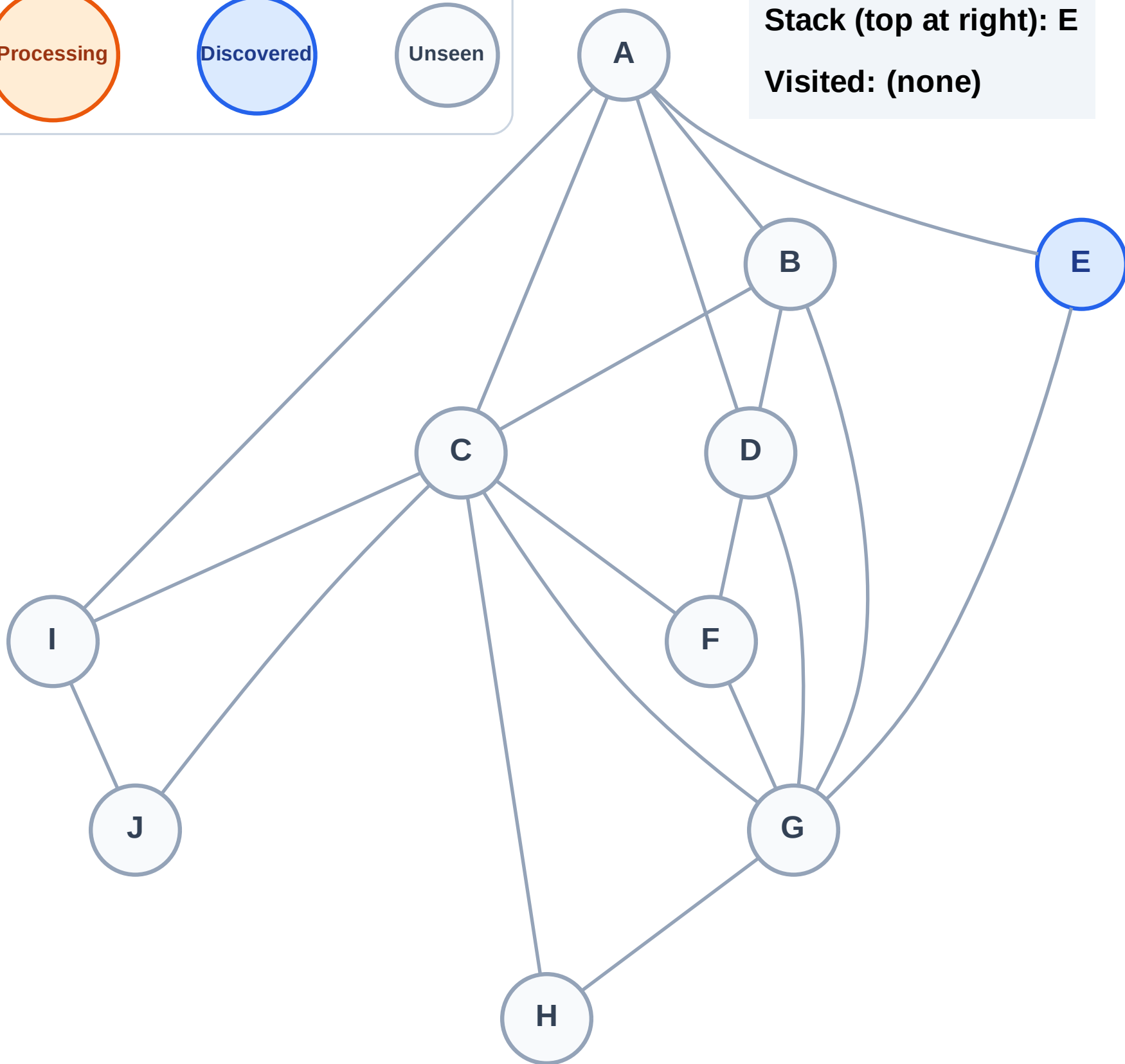
Step 1: Start at E

## Legend



Stack (top at right): E

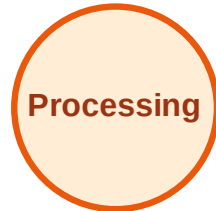
Visited: (none)



# DFS Traversal

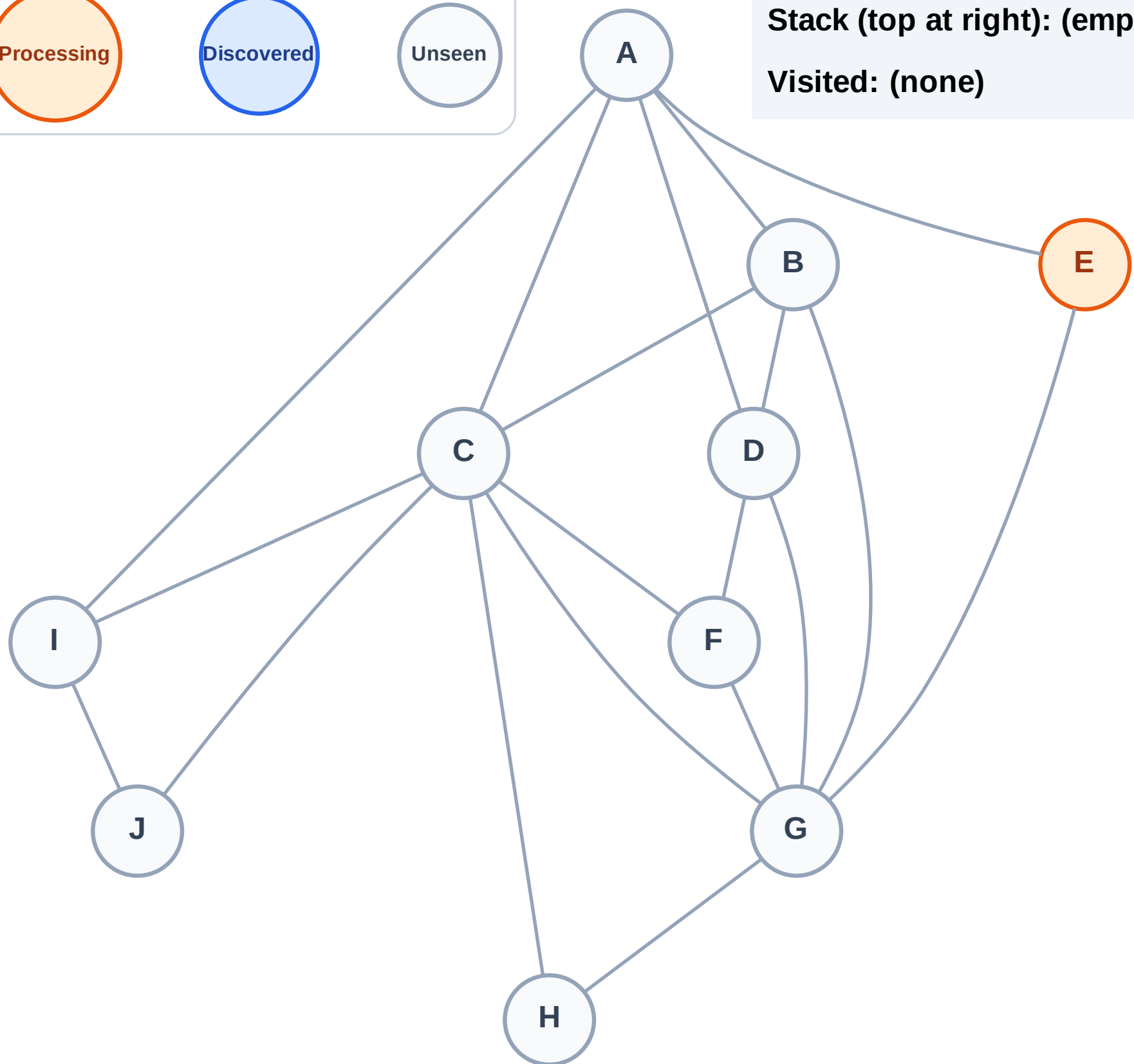
Step 2: Process node E

## Legend



Stack (top at right): (empty)

Visited: (none)





# DFS Traversal

Step 3: Discover G, A from E

Legend

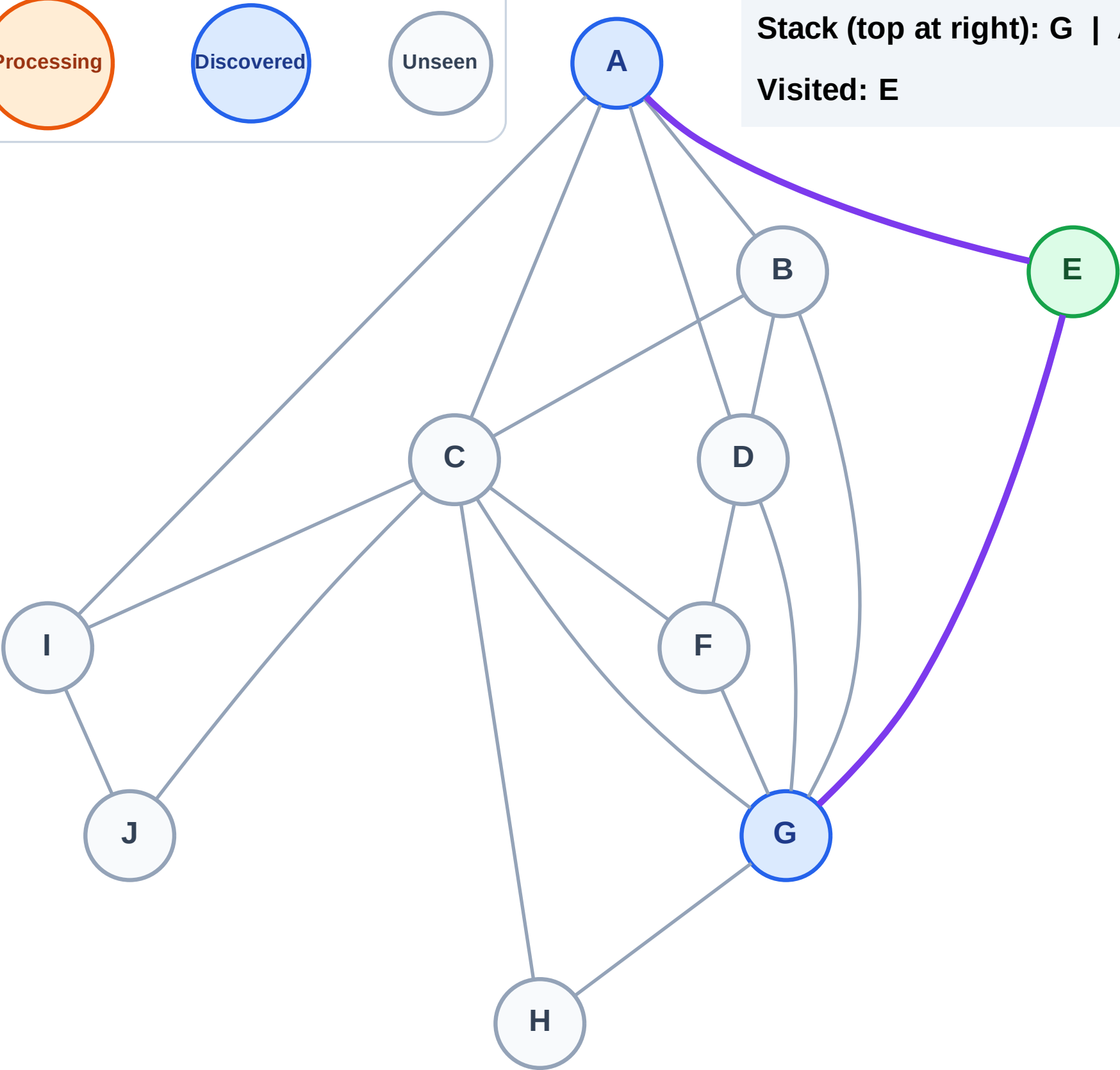
Complete

Processing

Discovered

Unseen

Stack (top at right): G | A  
Visited: E





# DFS Traversal

Step 4: Process node A

Legend

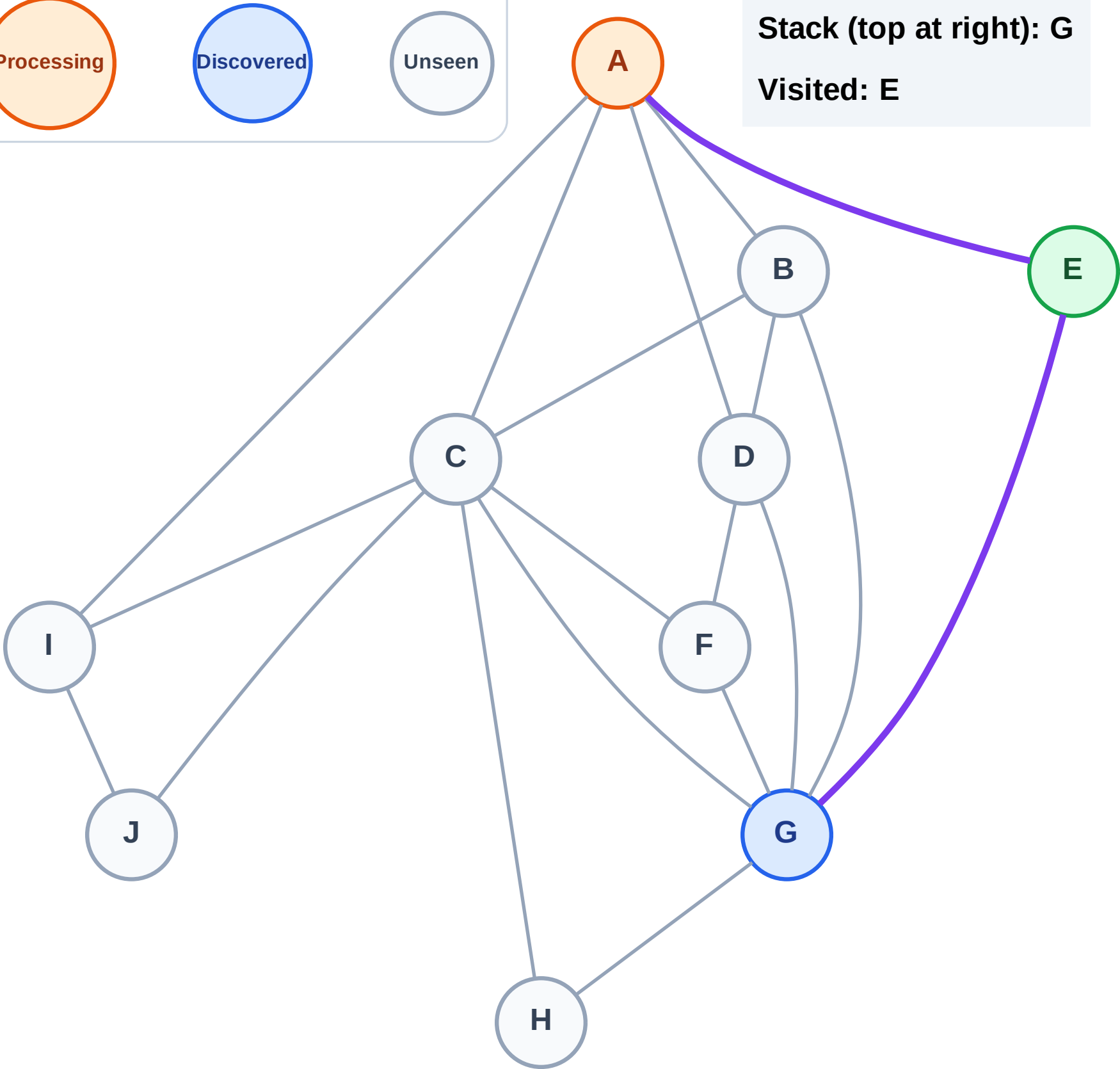
Complete

Processing

Discovered

Unseen

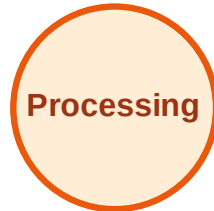
Stack (top at right): G  
Visited: E



# DFS Traversal

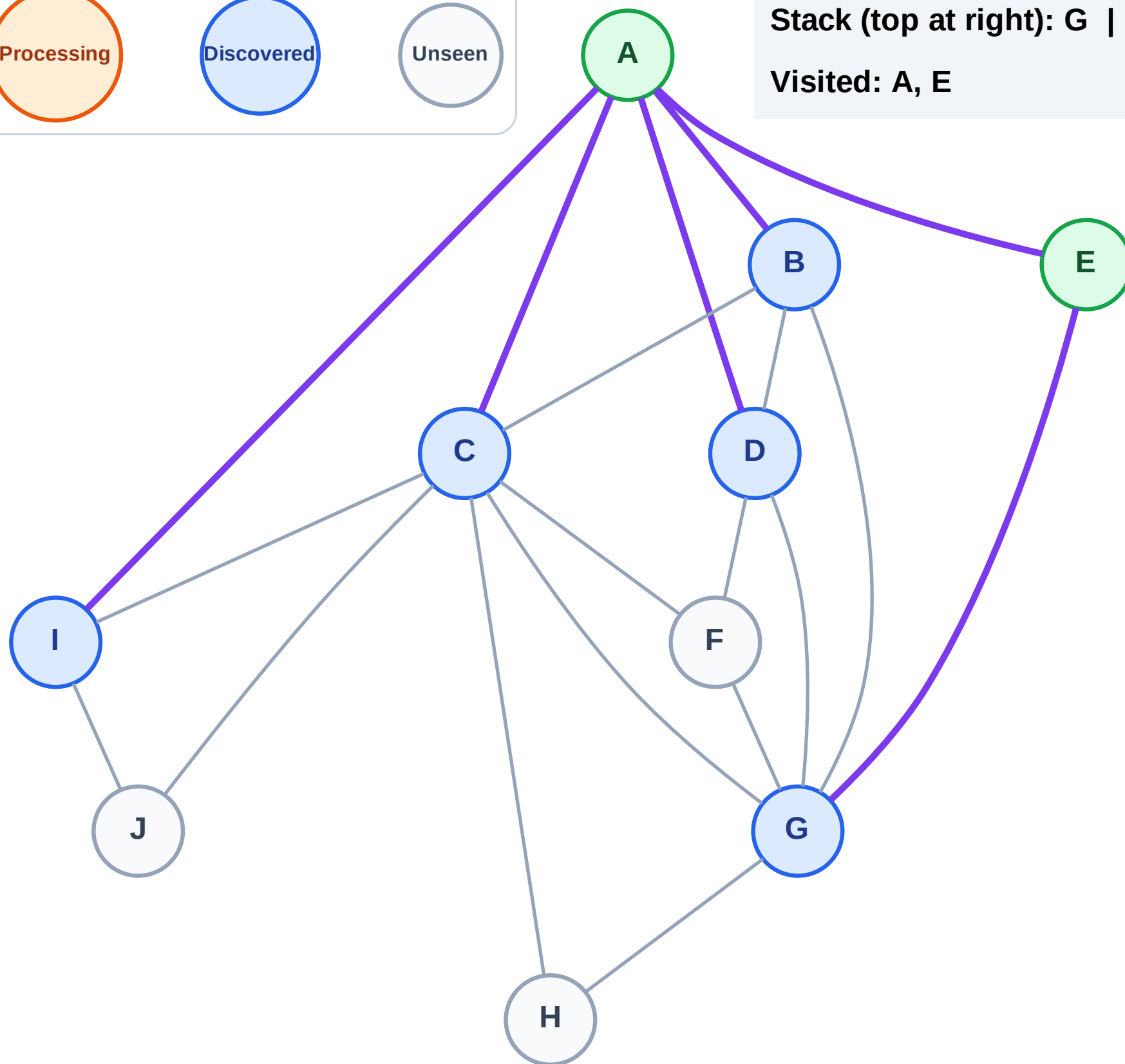
Step 5: Discover I, D, C, B from A

## Legend



Stack (top at right): G | I | D | C | B

Visited: A, E



# DFS Traversal

Step 6: Process node B

Legend

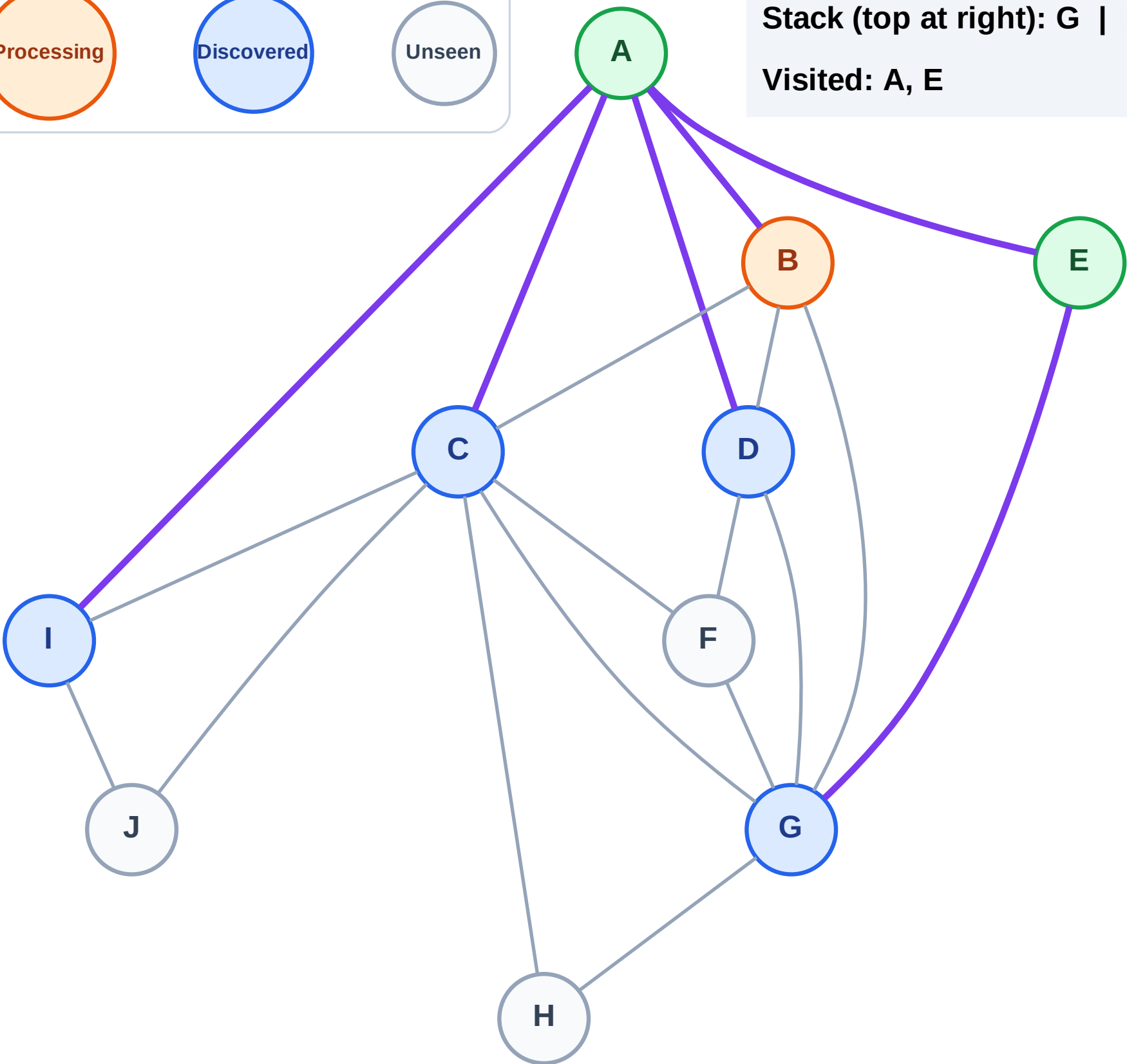
Complete

Processing

Discovered

Unseen

Stack (top at right): G | I | D | C  
Visited: A, E



# DFS Traversal

Step 7: No new neighbors from B

Legend

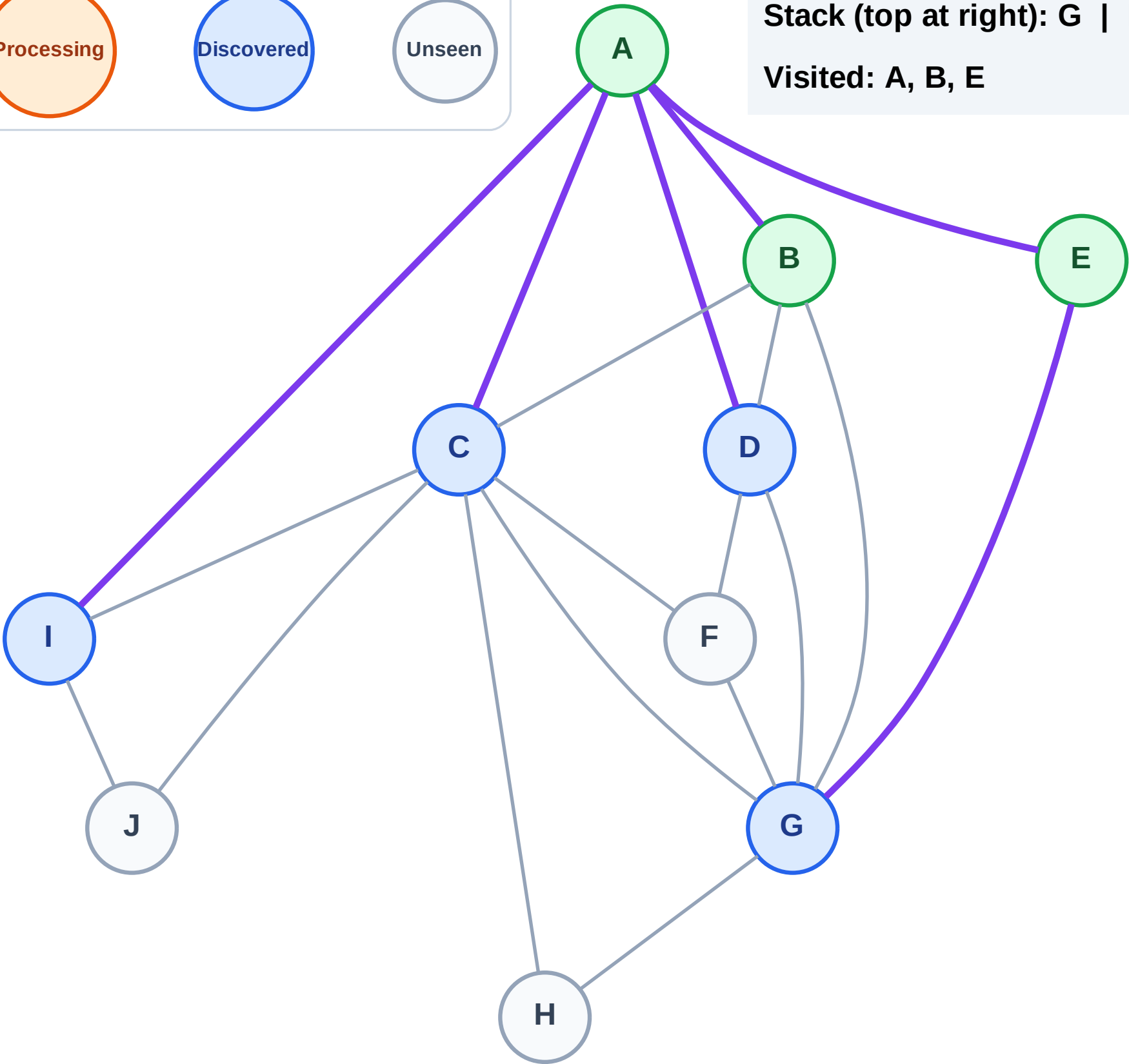
Complete

Processing

Discovered

Unseen

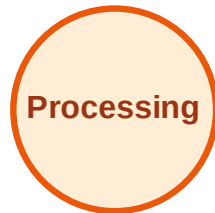
Stack (top at right): G | I | D | C  
Visited: A, B, E



# DFS Traversal

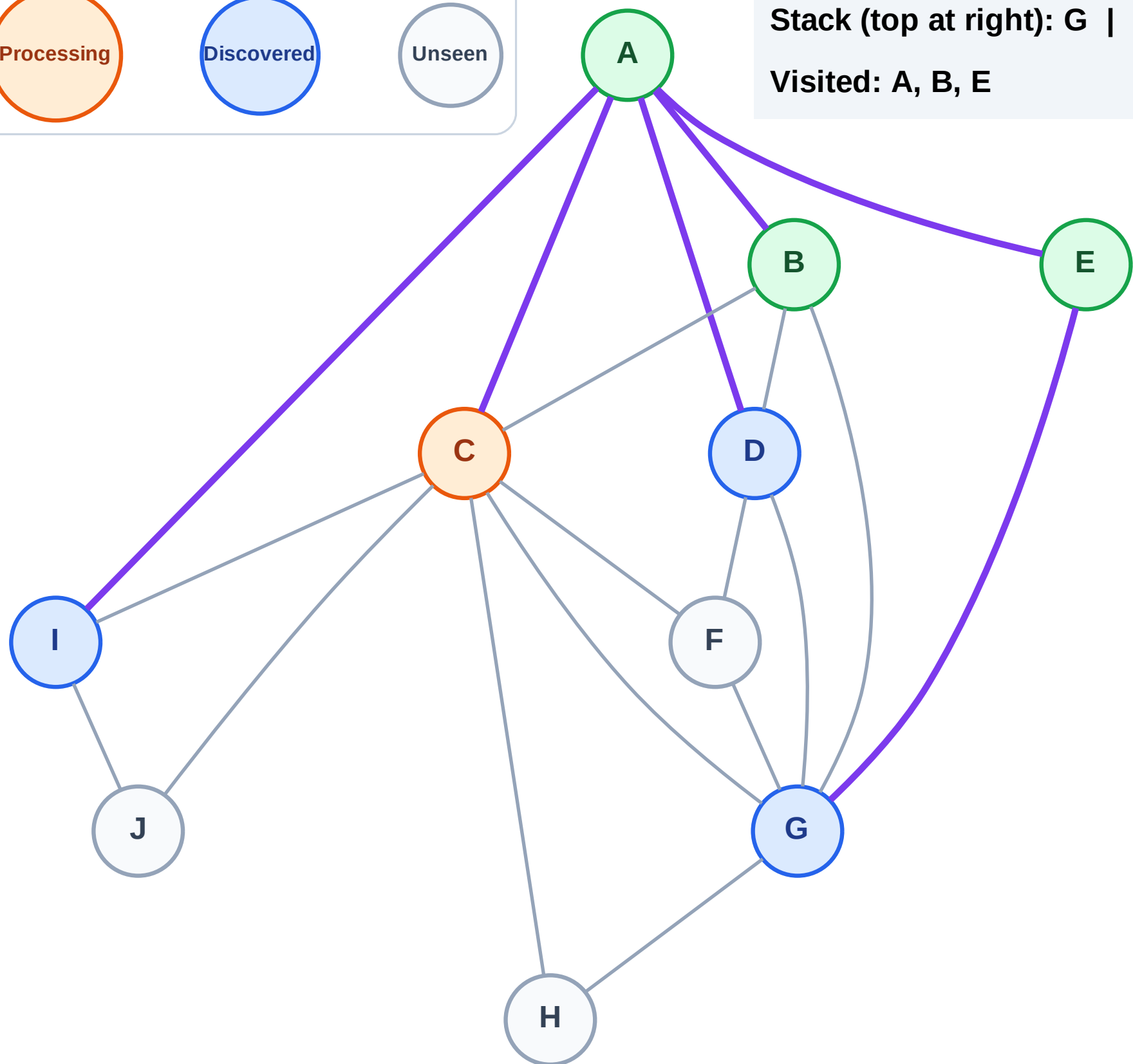
Step 8: Process node C

## Legend



Stack (top at right): G | I | D

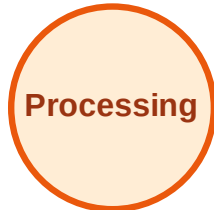
Visited: A, B, E



# DFS Traversal

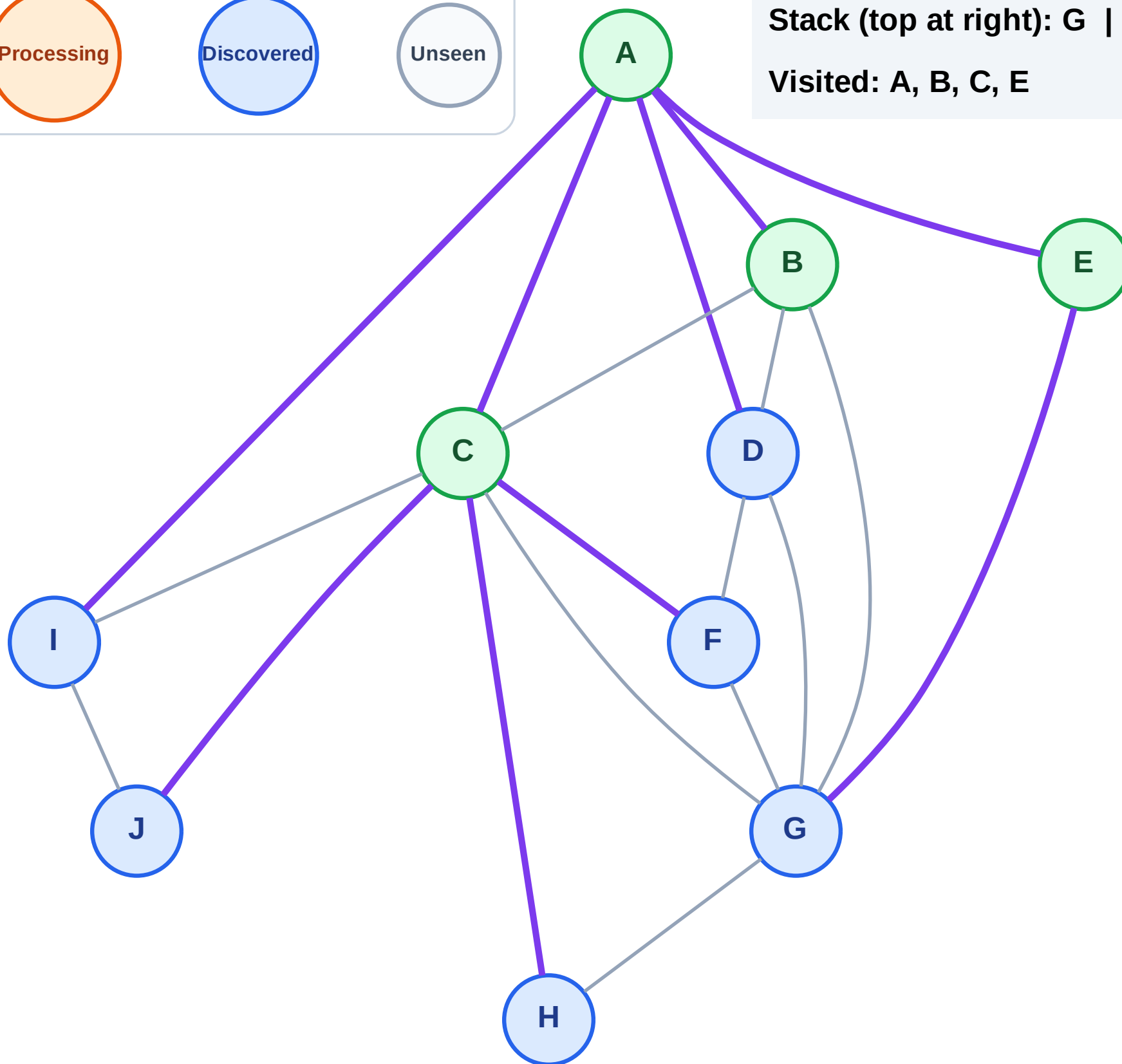
Step 9: Discover J, H, F from C

## Legend



Stack (top at right): G | I | D | J | H | F

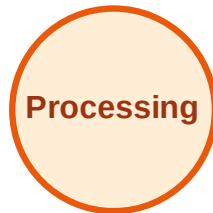
Visited: A, B, C, E



# DFS Traversal

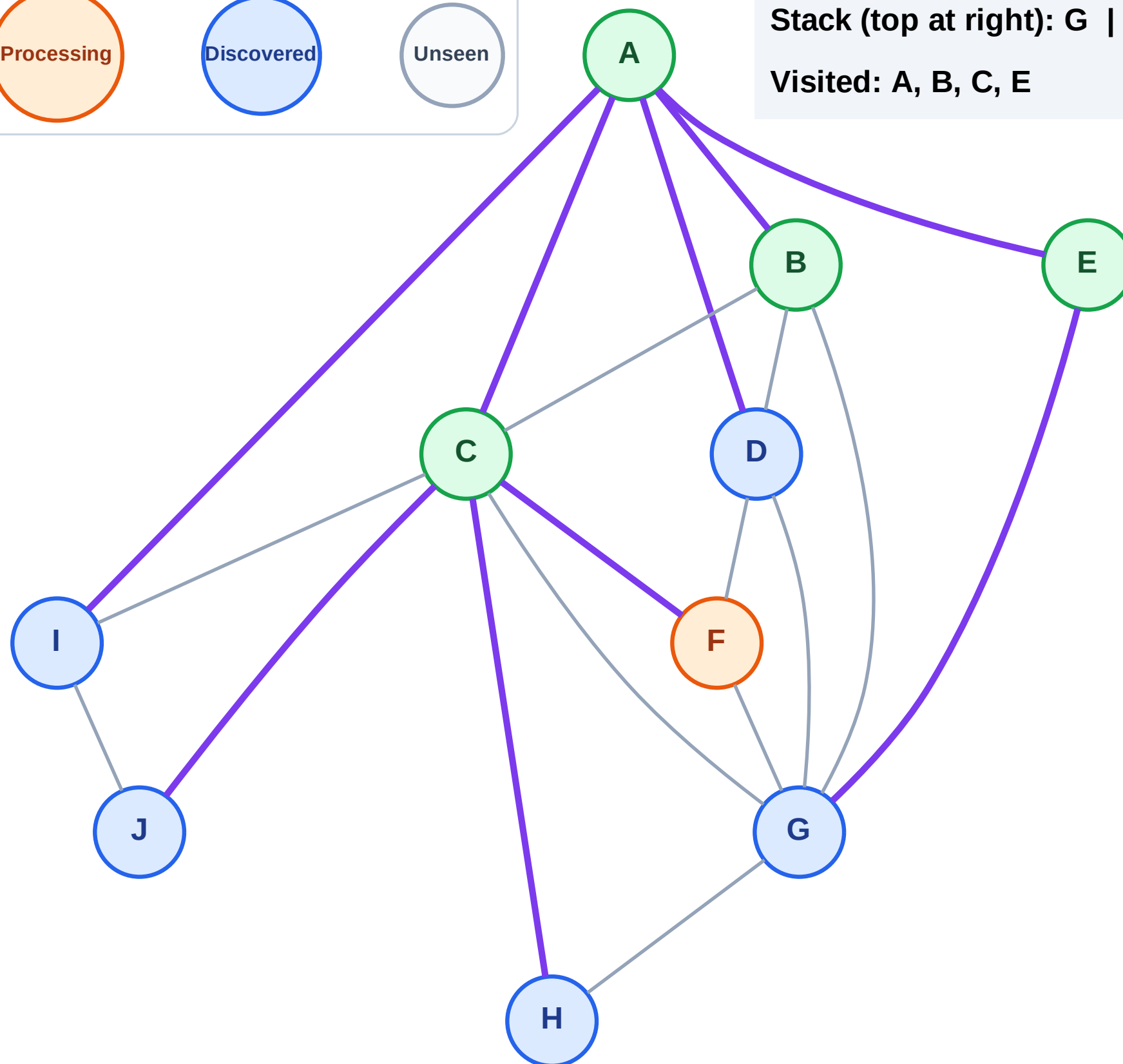
Step 10: Process node F

## Legend



Stack (top at right): G | I | D | J | H

Visited: A, B, C, E

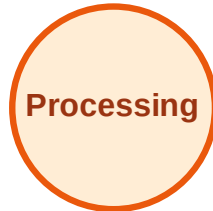




# DFS Traversal

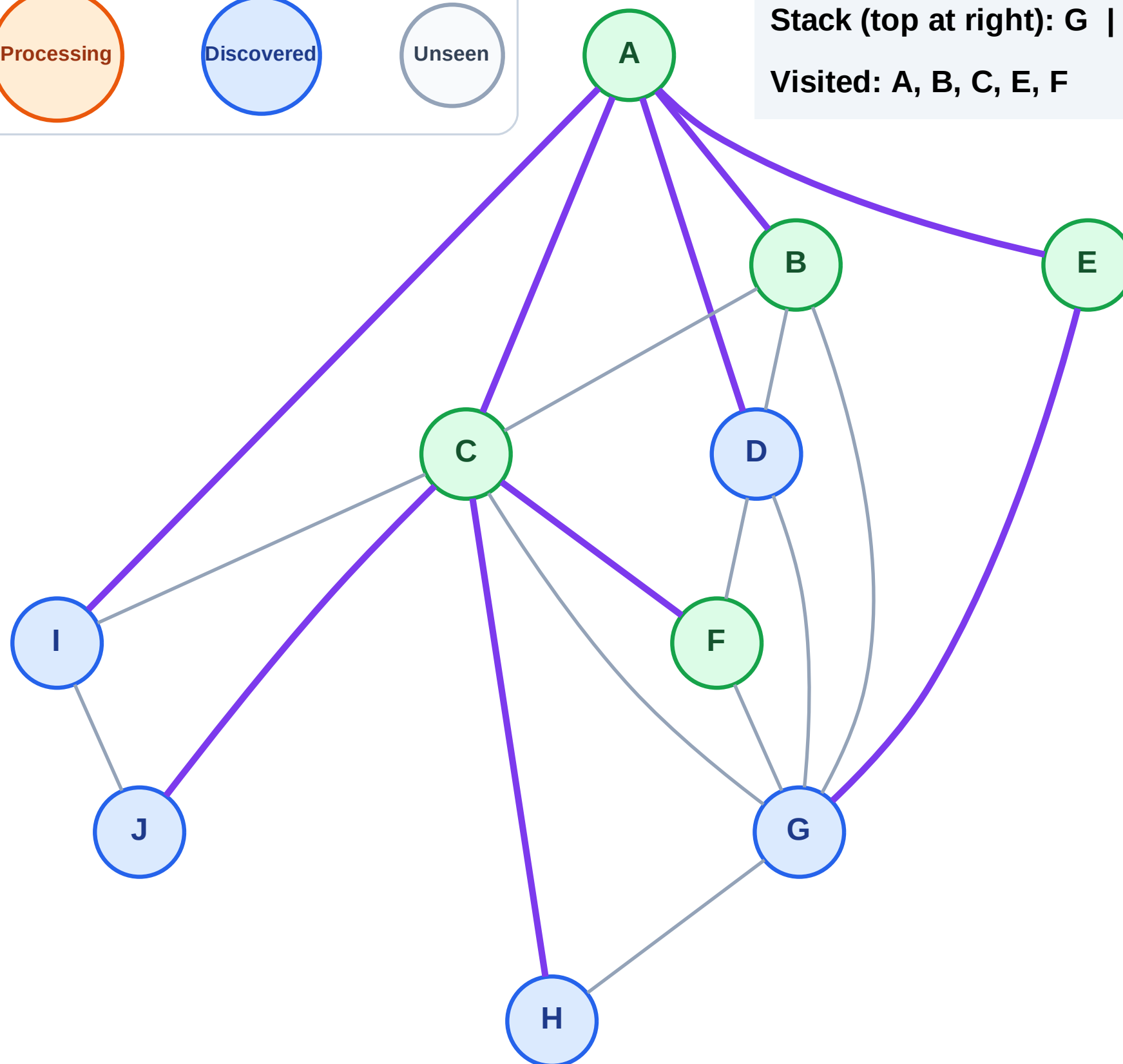
Step 11: No new neighbors from F

## Legend



Stack (top at right): G | I | D | J | H

Visited: A, B, C, E, F

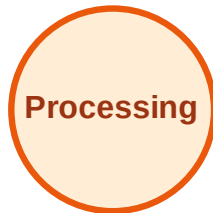




# DFS Traversal

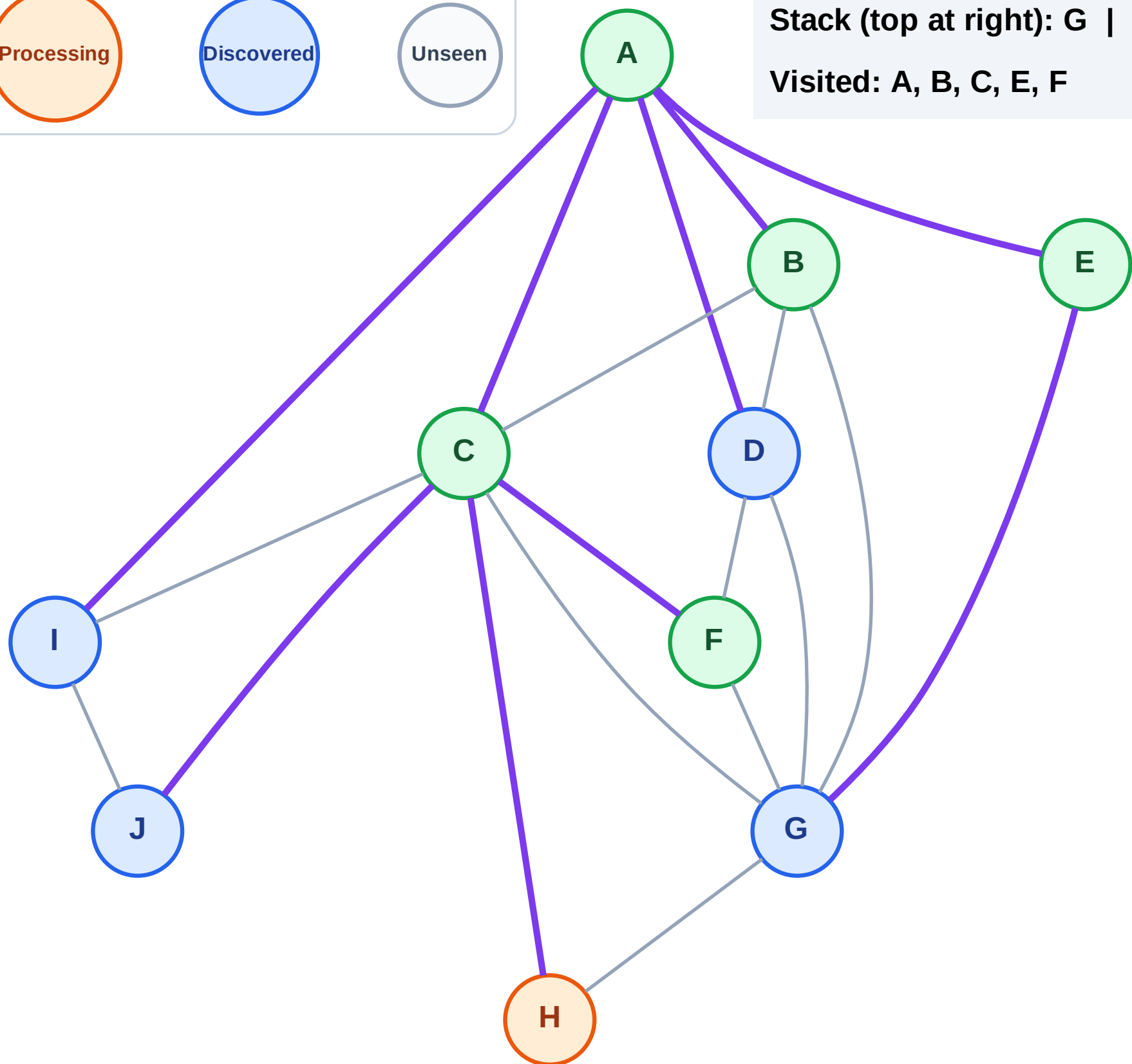
Step 12: Process node H

## Legend



Stack (top at right): G | I | D | J

Visited: A, B, C, E, F



# DFS Traversal

Step 13: No new neighbors from H

Legend

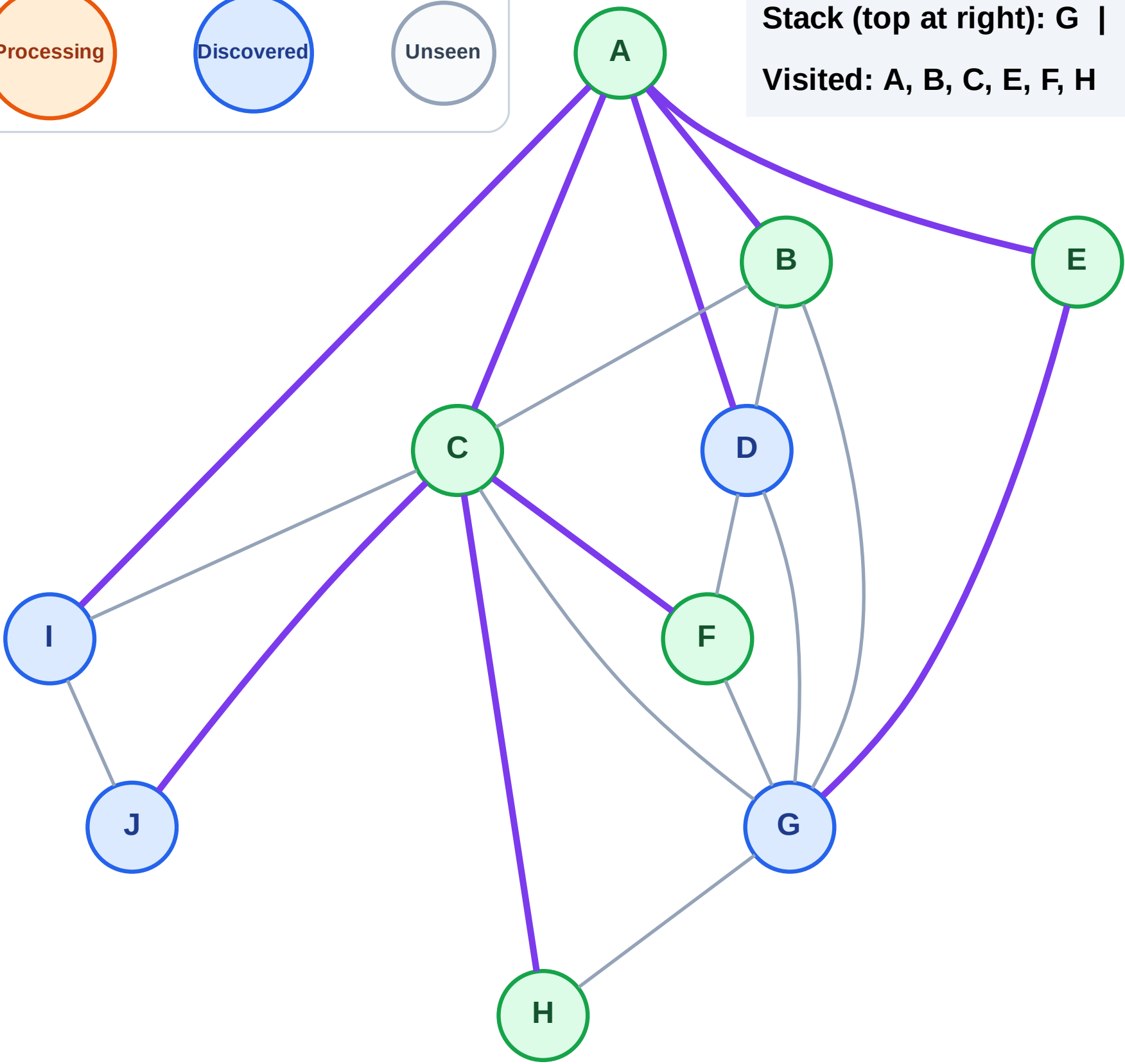
Complete

Processing

Discovered

Unseen

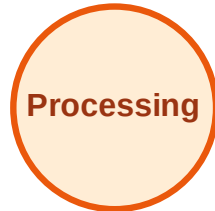
Stack (top at right): G | I | D | J  
Visited: A, B, C, E, F, H



# DFS Traversal

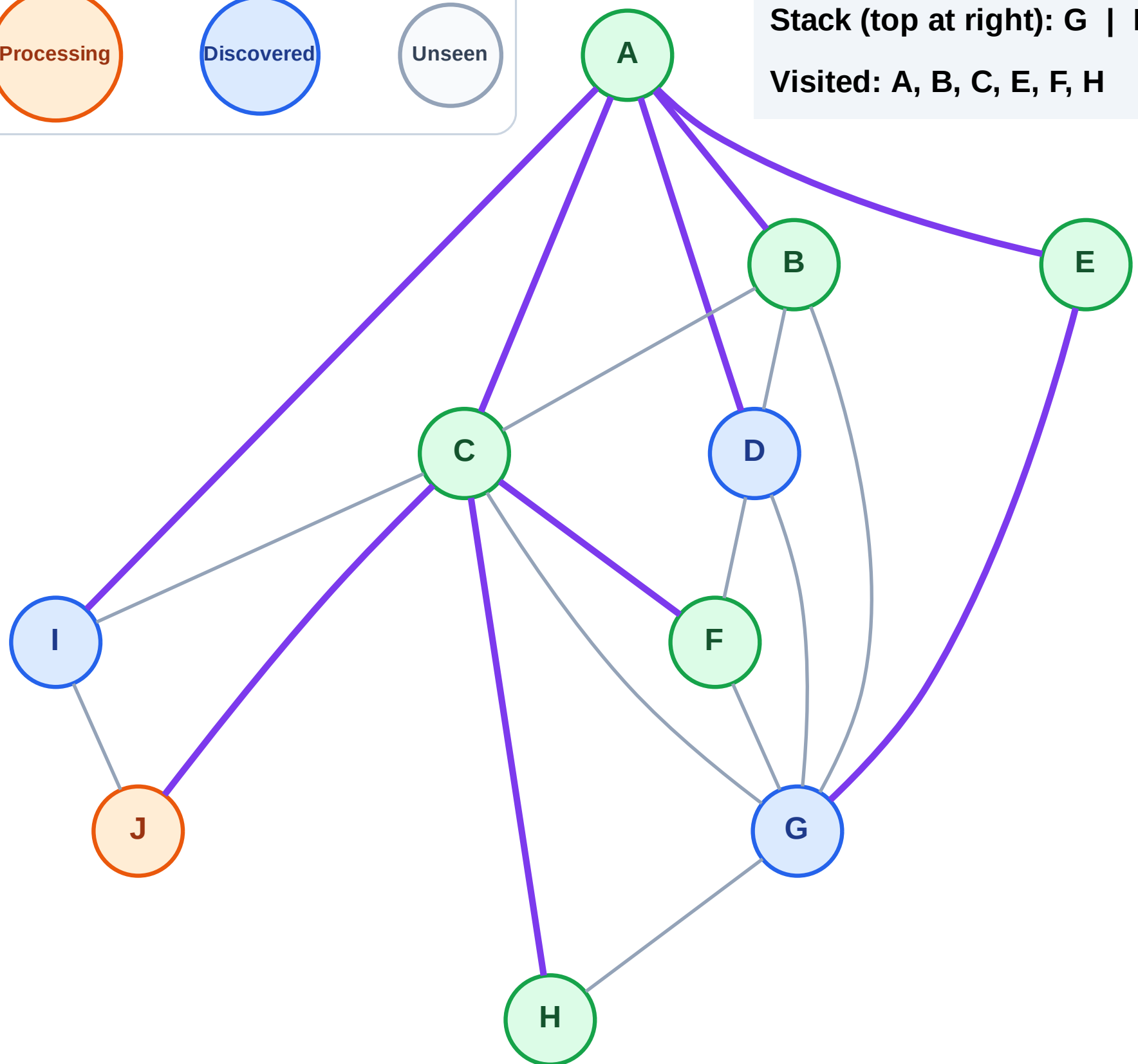
Step 14: Process node J

## Legend



Stack (top at right): G | I | D

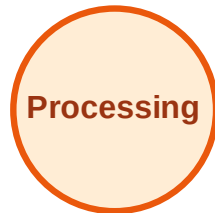
Visited: A, B, C, E, F, H



# DFS Traversal

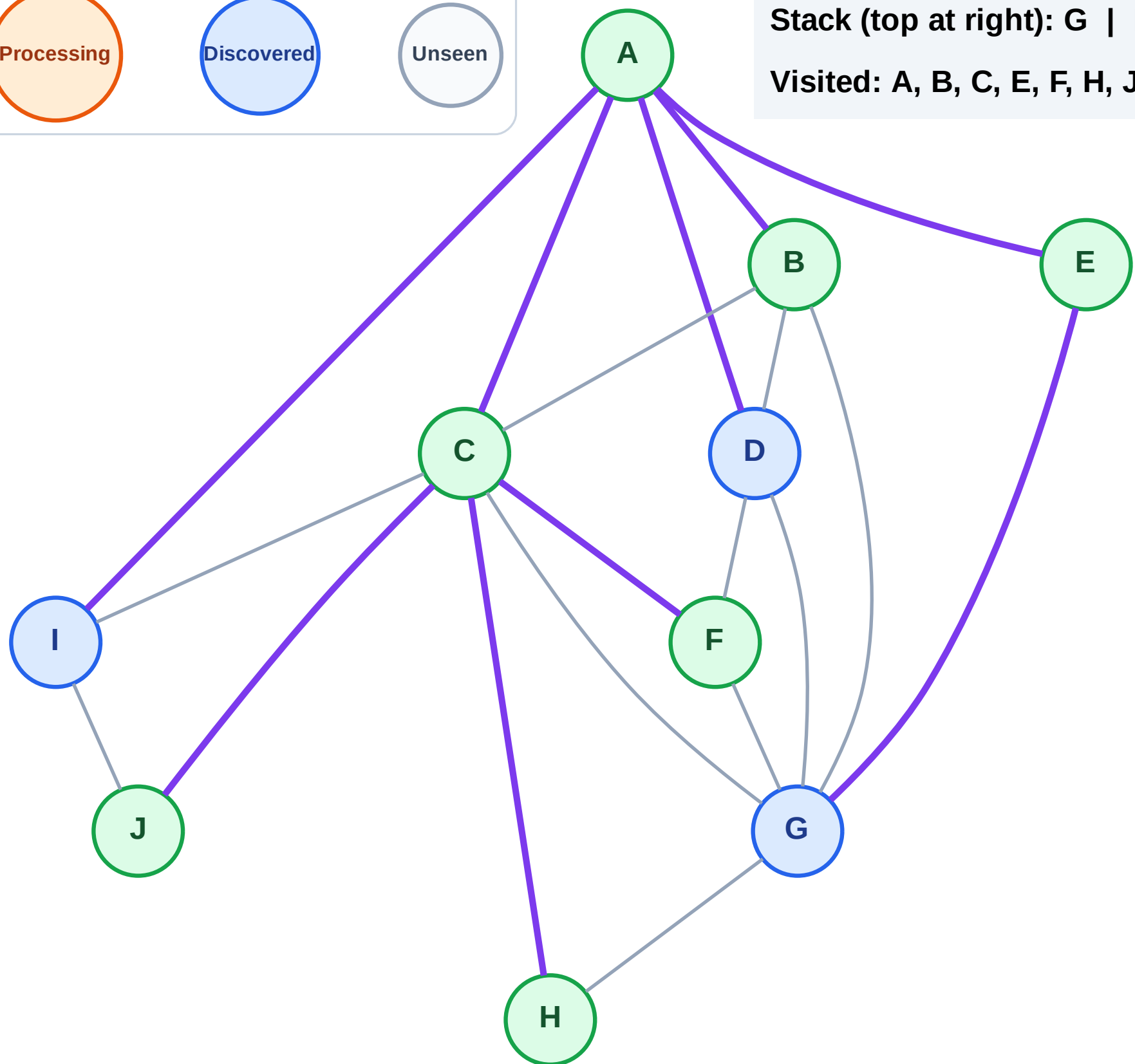
Step 15: No new neighbors from J

## Legend



Stack (top at right): G | I | D

Visited: A, B, C, E, F, H, J



# DFS Traversal

Step 16: Process node D

## Legend

Complete

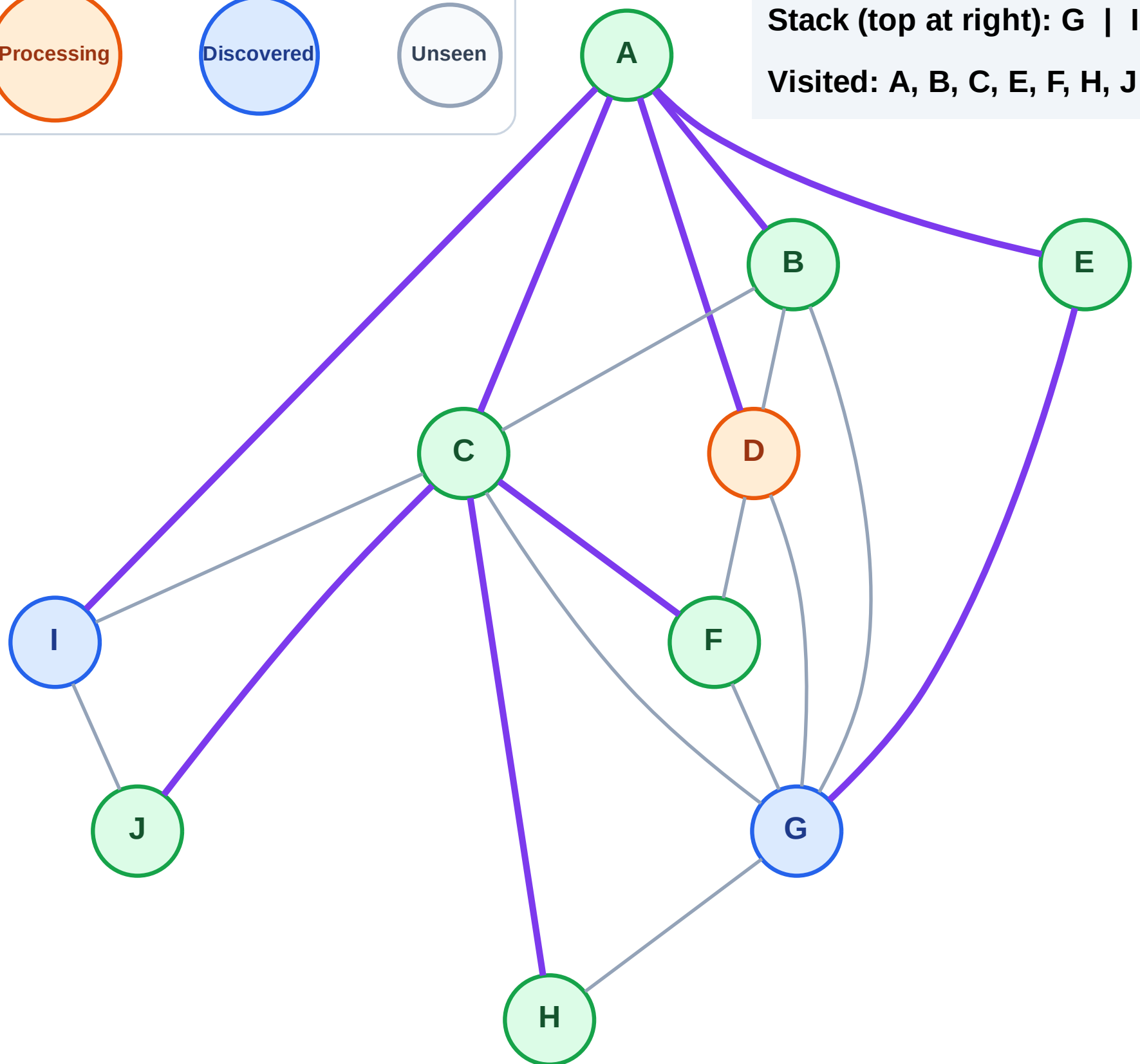
Processing

Discovered

Unseen

Stack (top at right): G | I

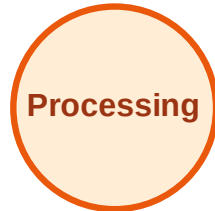
Visited: A, B, C, E, F, H, J



# DFS Traversal

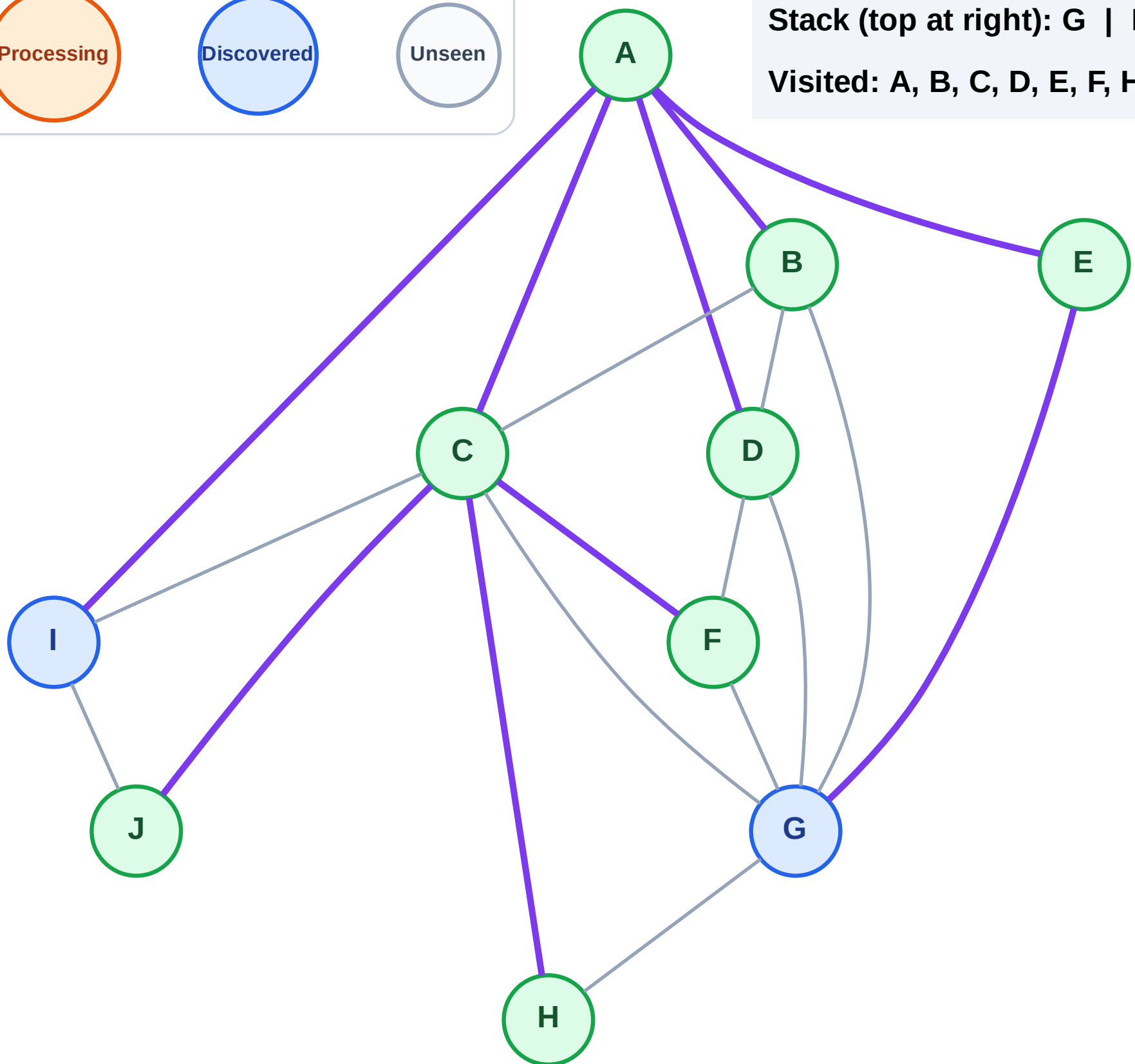
Step 17: No new neighbors from D

## Legend



Stack (top at right): G | I

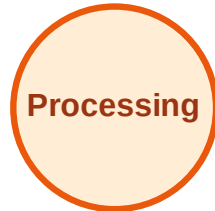
Visited: A, B, C, D, E, F, H, J



# DFS Traversal

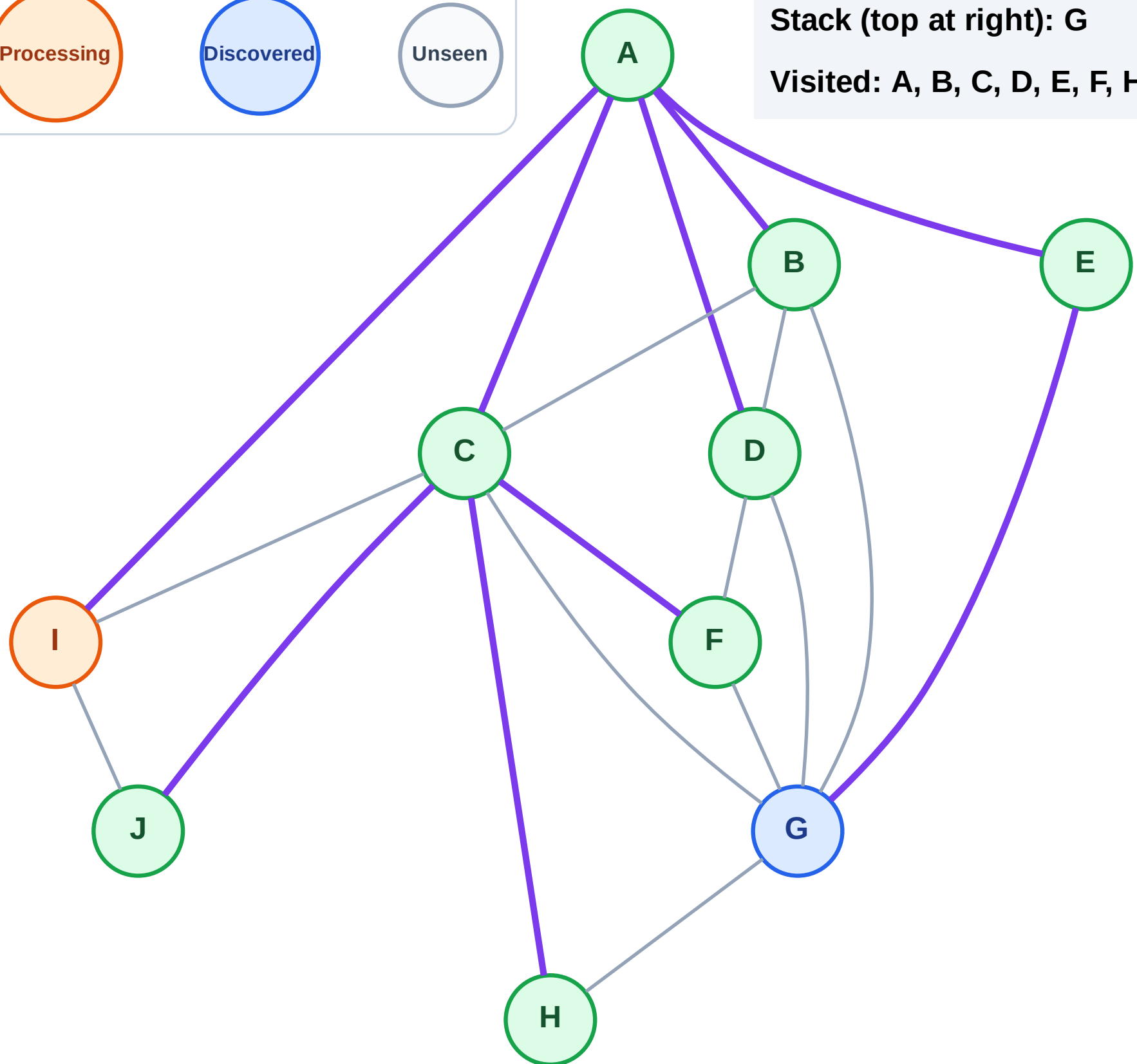
Step 18: Process node I

## Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, J

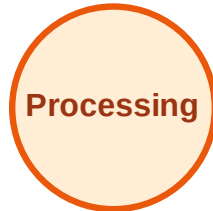




# DFS Traversal

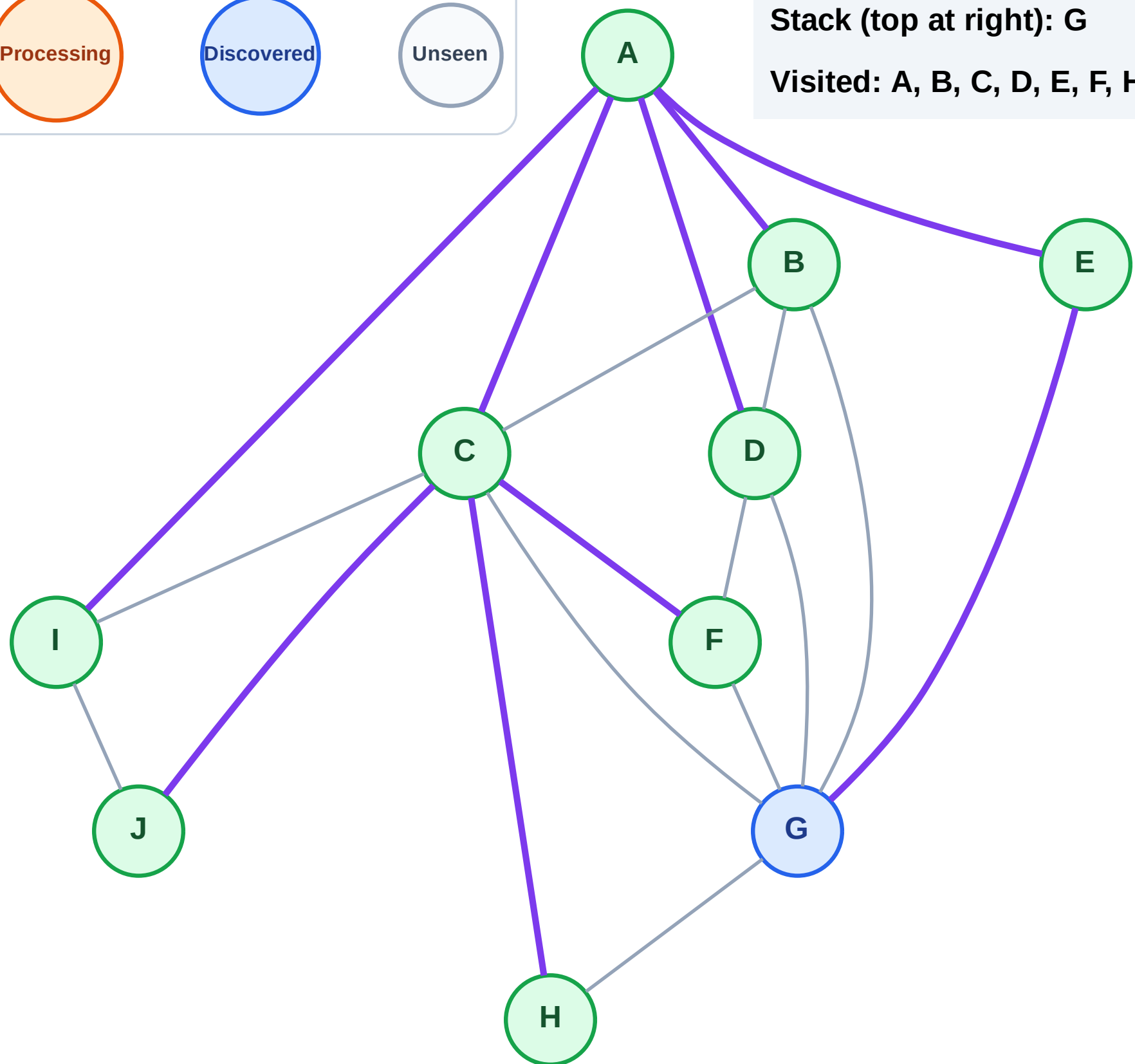
Step 19: No new neighbors from I

## Legend



Stack (top at right): G

Visited: A, B, C, D, E, F, H, I, J

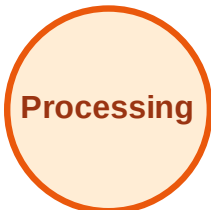




# DFS Traversal

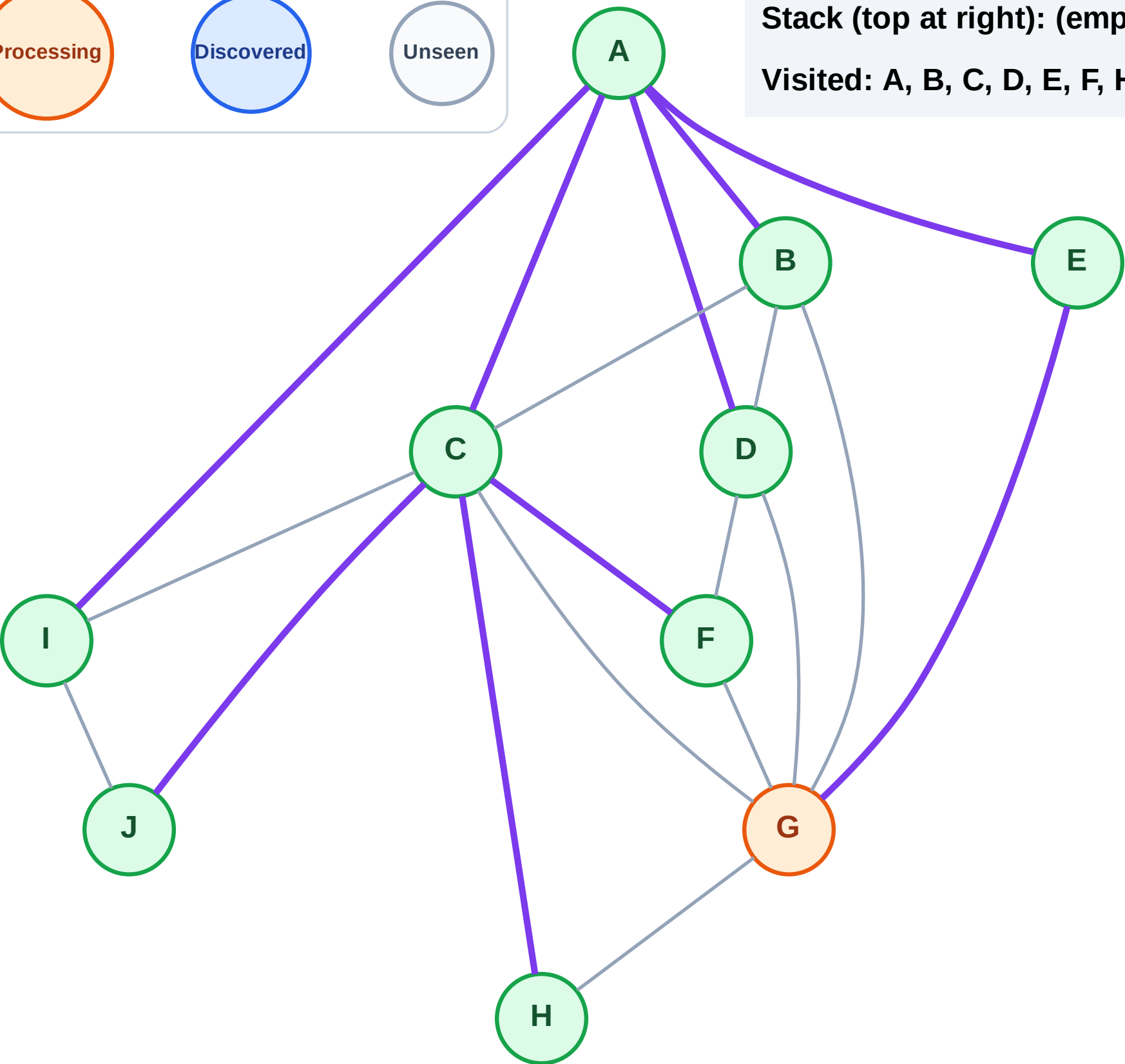
Step 20: Process node G

## Legend



Stack (top at right): (empty)

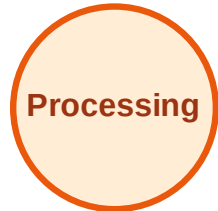
Visited: A, B, C, D, E, F, H, I, J



# DFS Traversal

Step 21: No new neighbors from G

## Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

